



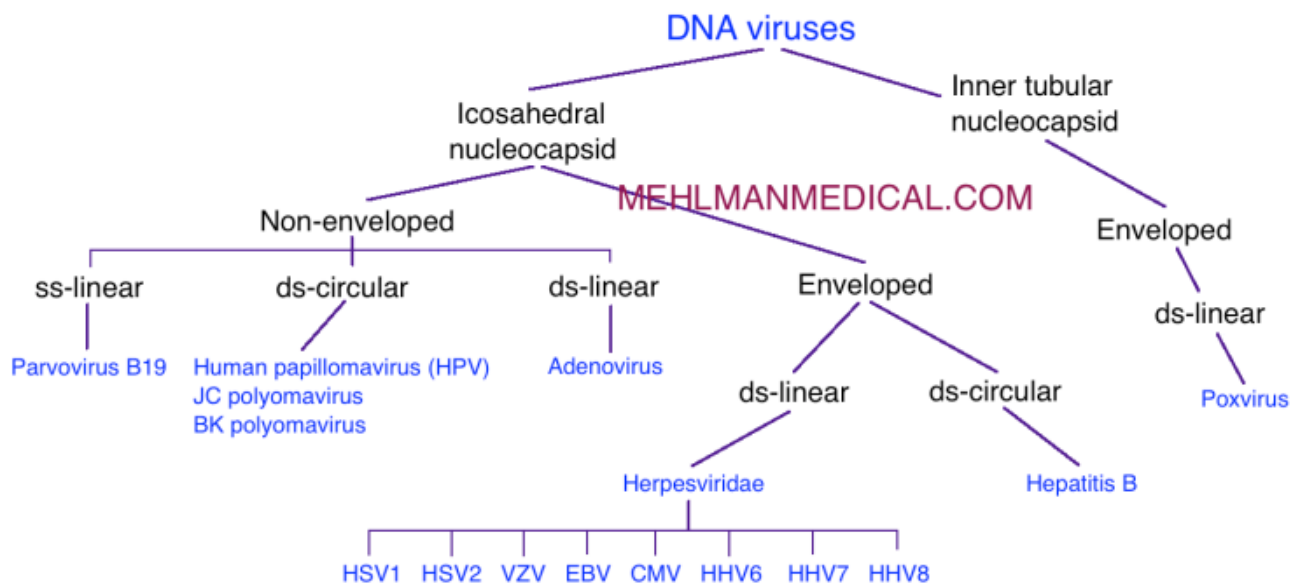
MEHLMANMEDICAL
MICROBIOLOGY
ASSESSMENT #1

Microbiology Assessment #1 – Virology:

1. A 22-year-old female presents to the physician with a painful vesicular eruption on her left labium majus. She describes the pain as burning and stinging. She has a fever of 101F and left-sided inguinal lymphadenopathy. Which of the following best describes the viral etiology?

- A) DNA; enveloped; linear
- B) DNA; enveloped; circular
- C) DNA; non-enveloped; linear
- D) DNA; non-enveloped; circular
- E) RNA; enveloped; linear
- F) RNA; enveloped; circular
- G) RNA; non-enveloped; linear
- H) RNA; non-enveloped; circular

The answer is A.



This patient has an acute genital herpes infection.

This may be due to either HSV1 or HSV2. The literature has in the past supported a propensity for HSV1 to cause herpes labialis (oral herpes) and HSV2 genital herpes, but the relationship is not absolute and considerable overlap exists.^{1, 2, 3}

Primary herpes infection

- Primary (first-time) infection often presents more severe than subsequent flareups, and is frequently accompanied by fever and regional adenopathy.⁴
- Systemic symptoms, including headache, fever, myalgia, and backache occur in about 70% of women and 40% of men.^{4, 5}

The herpesviridae are double-stranded **linear**, enveloped, DNA viruses.⁶

This is in contrast to hepatitis B, which is a double-stranded **circular**, enveloped, DNA virus.⁷

The USMLE really likes this distinguishing structural detail between herpesviridae and hepatitis B.

Bottom line: On your USMLE: Herpes is “DNA, enveloped, linear.” Hepatitis B is “DNA, enveloped, circular.”

1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2564733/>

2) <https://www.ncbi.nlm.nih.gov/pubmed/62191/>

- 3) <https://www.ncbi.nlm.nih.gov/pubmed/204741/>
- 4) <https://www.ncbi.nlm.nih.gov/books/NBK47449/>
- 5) <https://www.ncbi.nlm.nih.gov/pubmed/6344712>
- 6) <https://www.ncbi.nlm.nih.gov/pubmed/28528442>
- 7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3940099/>

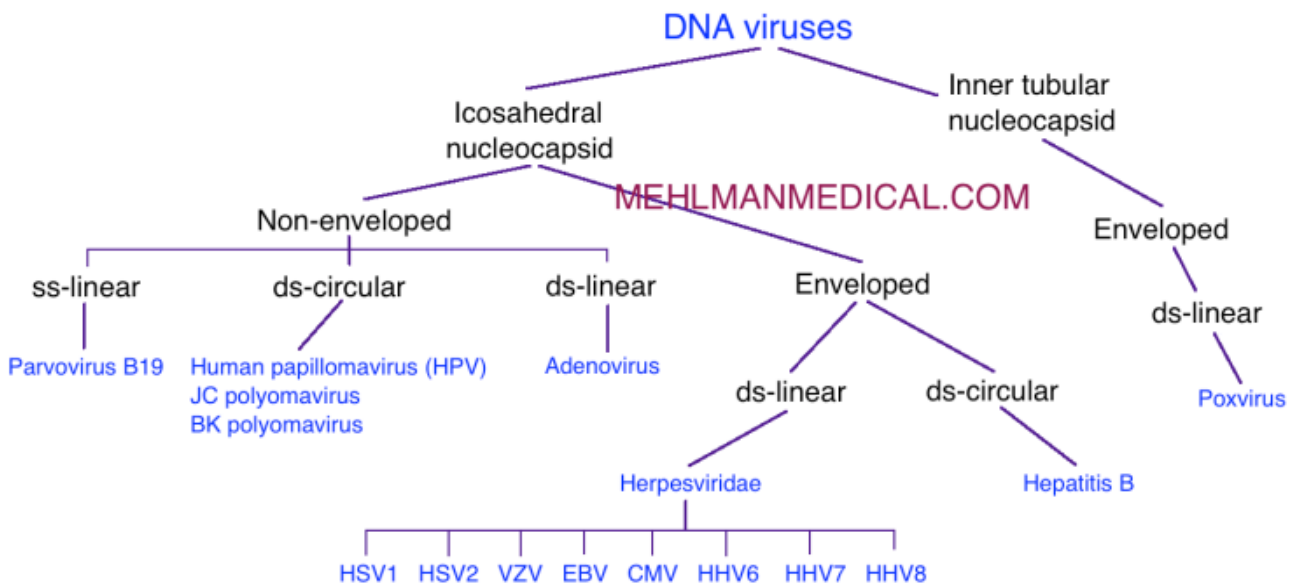
2.



A 9-year-old boy presents with the above lesions on his trunk after returning home from a one-month-long summer camp. Which of the following might be seen under light microscopy if a biopsy is taken of one of the lesions?

- A) Multinucleated T lymphocytes
- B) Intranuclear inclusions
- C) Large intracytoplasmic, basophilic inclusions
- D) Small- or medium-sized lymphocytes with cerebriform nuclei

The answer is C.



This patient has **molloscum contagiosum** (MC) caused by **poxvirus**.

Molloscum contagiosum^{1,2}

- Presents usually as shiny, dome-shaped, peach-colored papules, with a central umbilication.
- Normally presents in children.
- Transmission classically associated with sharing towels (e.g., at public swimming pools) and exposure to contaminated fomites (inanimate objects).
- Lesions appear 2-6 weeks after viral exposure and disappear on their own over the course of months to a few years, with an average duration of about one year.

Biopsy of lesions shows **Molloscum bodies**, which are large, intracytoplasmic, basophilic inclusions within keratinocytes.^{3,4}

Sounds pedantic, but Molloscum bodies are HY on the USMLE.

“Multinucleated T lymphocytes” refers to Warthin-Finkeldey cells, which can be harvested from the tonsils of patients in the prodromal stage of measles.⁵

“Intranuclear inclusions” refers to Cowdry bodies, which are associated with herpesviridae infections.⁶

“Small- or medium-sized lymphocytes with cerebriform nuclei” refers to mycosis fungoides and Sezary syndrome, which are cutaneous T-cell lymphoma and leukemia, respectively. These conditions are associated with human T-cell lymphotropic virus (HTLV).^{7,8}

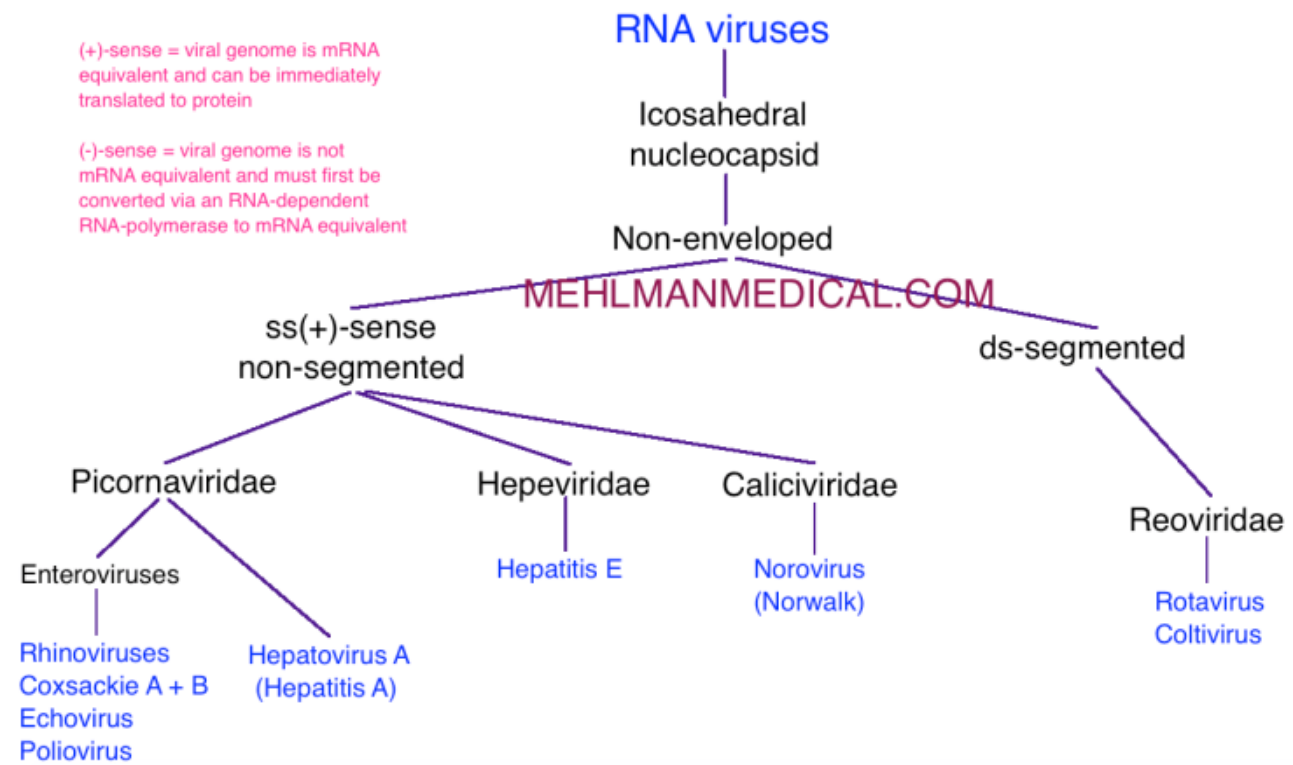
Bottom line: Poxvirus causes molluscum contagiosum. It classically presents in children as shiny, dome-shaped, peach-colored papules, with a central umbilication. Biopsy shows molluscum bodies, which are large, intracytoplasmic, basophilic inclusions within keratinocytes.

- 1) <https://jamanetwork.com/journals/jamadermatology/fullarticle/2547245>
- 2) <https://www.cdc.gov/poxvirus/molluscum-contagiosum/transmission.html>
- 3) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3687174/>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3001182/>
- 5) <https://academic.oup.com/ajcp/article-abstract/97/2/179/1779694>
- 6) <https://journals.lww.com/ajsp/abstract/1989/02000/>
- 7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC295510/>
- 8) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3699909/>

3. A 3-year-old boy is brought in to the GP by his mother for severe watery diarrhea the past two days. He appears pale and his orbits are sunken. His history is significant for not having received childhood vaccinations. Which of the following best describes the most likely viral etiology?

- A) RNA; enveloped; single-stranded; non-segmented
- B) RNA; enveloped; single-stranded; segmented
- C) RNA; enveloped; double-stranded; non-segmented
- D) RNA; enveloped; double-stranded; segmented
- E) RNA; non-enveloped; single-stranded; non-segmented
- F) RNA; non-enveloped; single-stranded; segmented
- G) RNA; non-enveloped; double-stranded; non-segmented
- H) RNA; non-enveloped; double-stranded; segmented

The answer is H.



This child has **rotavirus**, which is the world's leading cause of severe, dehydrating diarrhea in children under age 5.^{1,2} Older children and adults can also experience infection but not to the same degree of prevalence as in young children.^{3,4,5}

Symptoms usually start two days after fecal-oral exposure. Vomiting and watery diarrhea tend to last 3-8 days.⁶

Rotavirus vaccine is administered orally at age 2, 4, and 6 **months** (3 doses total).⁷

On the USMLE, if you get a vignette of a young child with watery diarrhea and they don't mention anything about vaccination history, rotavirus is the answer.

If the USMLE question specifically mentions that the child was vaccinated or that the parents and/or siblings *also* simultaneously have diarrhea, then rotavirus is less likely. **Norwalk virus (Norovirus)** is the second-leading cause of diarrhea in children after rotavirus⁸, and it is also common in older children and adults.⁹

The USMLE wants you to know that rotavirus is a double-stranded segmented virus (11 segments).¹⁰

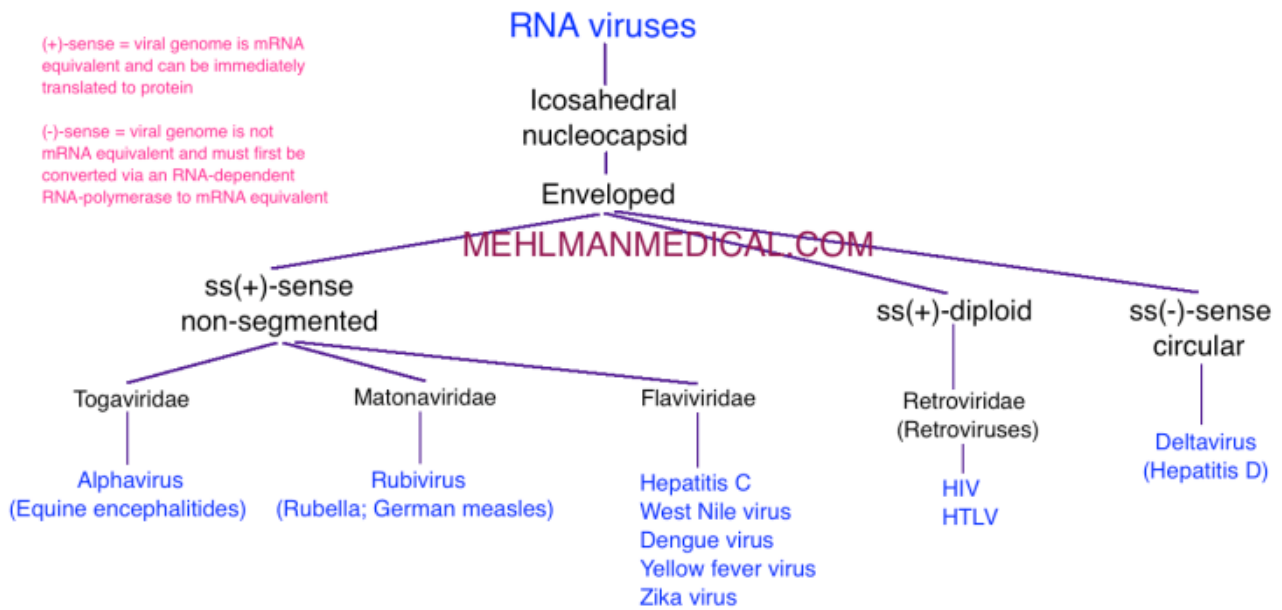
Bottom line: Rotavirus is the most common cause of diarrhea in children under age 5; it is *less common* in older children and adults. Norwalk virus (Norovirus) is the second-leading cause of diarrhea in young children; it is *common* in older children and adults.

- 1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5858916/>
- 2) <https://www.who.int/immunization/diseases/rotavirus/en/>
- 3) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4017201/>
- 4) <https://www.ncbi.nlm.nih.gov/pubmed/14871633>
- 5) <https://www.ncbi.nlm.nih.gov/pubmed/28067093/>
- 6) <https://www.cdc.gov/rotavirus/about/symptoms.html>
- 7) <https://www.cdc.gov/vaccines/vpd/rotavirus/index.html>
- 8) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3303077/>
- 9) <https://www.cdc.gov/norovirus/about/symptoms.html>
- 10) <https://www.ncbi.nlm.nih.gov/pubmed/21195621>

4. A 6-year-old girl has a fever of 101F, malaise, and anorexia for two days. As the fever abates, a maculopapular rash starts on her face and spreads downward to involve her entire body; it lasts three days. On examination she has tenderness to palpation of the suboccipital and postauricular areas. Which of the following is the most likely diagnosis?

- A) Rubeola
- B) German measles
- C) Mumps
- D) Roseola
- E) Fifth disease

The answer is B.



Rubella (German measles; three-day measles)

Presentation¹

- Classically pediatric viral illness, with peak incidence age 5-9.
- Three-day prodrome of fever, malaise, and anorexia.
- As the fever abates, a maculopapular rash starts at the head and descends to cover the whole body.
- **Suboccipital and/or postauricular lymphadenopathy (exceedingly HY)**
- **Adults may present with arthritis**

Vaccine

- Rubella is rare in the United States because of the Mumps-Measles-Rubella (MMR) live-attenuated vaccine.¹
- First dose given at age 12-15 months; second dose given age 4-6 years.²
- MMR and varicella/zoster are the only live-attenuated vaccines approved for use in HIV patients (CD4 must be >200 cells/mm³).³

Congenital rubella¹

- Disease occurs secondary to first-trimester *in utero* infection in non-immune mother.
- As mentioned above, pregnant adult female may present **with arthritis**.
- Neonatal deafness, meningoencephalitis, cataracts
- **Patent ductus arteriosus (PDA) in neonate (exceedingly HY)**

Rubeola (measles)

- High fever, cough, coryza, conjunctivitis (3Cs).⁴
- **Koplik spots** 2-3 days after symptoms begin and are described as punctate white spots on an erythematous background on the buccal mucosa.⁵
- As the fever abates, a maculopapular rash starts at the head and descends to cover the whole body (similar to rubella).⁴

Mumps⁶

- Classically presents as **parotitis**; **orchitis** and **meningitis** may also be seen (**POM**).
- Rash is not part of the classic disease presentation.

Roseola infantum (exanthem subitum; Sixth disease)⁷

- Caused by HHV6 (human herpes virus 6)
- High-grade fever (i.e., up to 104F) for 3-5 days.
- This is followed by a rapid defervescence of the fever with an accompanying maculopapular rash.
- **For the USMLE, just remember: “spiking fever followed by a rash.”**

Fifth disease (erythema infectiosum; Fifth disease)⁸

- Characteristic facial rash described as a “slapped cheek” appearance
- By the time the rash appears, the virus is no longer infectious (i.e., cleared by the immune system). For behavioral science USMLE questions, you can tell the worried parent that the child will be okay.
- **Can cause aplastic anemia** (↓ RBCs + ↓ WBCs + ↓ platelets – i.e., all are decreased); more common in children with RBC conditions associated with shortened lifespan, i.e., **sickle cell**.
- Associated with daycare centers.
- Can present as arthritis and lacy rash in adult.

Bottom line: These conditions are known as viral exanthems. They are all exceedingly HY on the USMLE.

1) <https://www.sciencedirect.com/science/article/pii/B9780323035064102895>

2) <https://www.cdc.gov/vaccines/vpd/rubella/index.html>

3) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4117268/>

4) <https://www.cdc.gov/measles/symptoms/signs-symptoms.html>

5) <https://www.sciencedirect.com/science/article/pii/B9780323045797001520>

6) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5215468/>

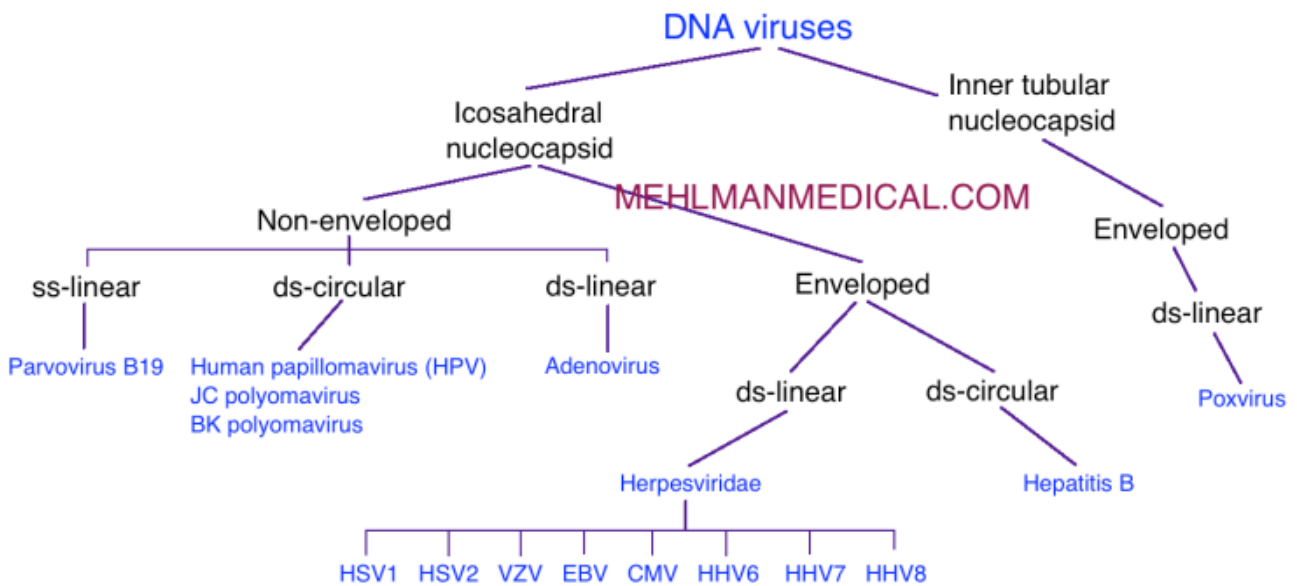
7) <https://www.ncbi.nlm.nih.gov/books/NBK448190/>

8) <https://www.aafp.org/afp/2007/0201/p373.html>

5. A 4-year-old who attends a daycare five days a week is brought to the GP by his mom for itchy, watery eyes. The conjunctivae are injected. The most likely causative virus is also known to cause which of the following?

- A) Hemorrhagic cystitis
- B) Uroepithelial cell tumorigenesis
- C) Hepatocellular neoplasia
- D) Retinal vascular proliferation
- E) Diabetes mellitus type I

The answer is A.



This patient has viral conjunctivitis.

Adenovirus is the most common cause of viral conjunctivitis and keratoconjunctivitis (inflammation of cornea + conjunctiva).^{1,2}

It is also associated with hemorrhagic cystitis, which presents as hematuria, dysuria, frequency, and urgency, usually resolving within two weeks.^{3,4}

Adenovirus is not believed to be associated with cancers in humans, although it may be used to transform mammalian cells *in vitro*.⁵

Coxsackie B virus⁶, rotavirus^{7,8}, mumps virus⁹, and cytomegalovirus are known viral precipitators of diabetes mellitus type I in individuals with susceptible HLA haplotypes¹⁰. Coxsackie B is the most common cause.¹¹

Bottom line: Adenovirus is the most common cause of viral conjunctivitis. It also causes hemorrhagic cystitis.

1) <https://www.ncbi.nlm.nih.gov/books/NBK470271/>

2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3086413/>

3) <https://jamanetwork.com/journals/jamapediatrics/article-abstract/503964>

4) <https://academic.oup.com/cid/article/32/9/1325/291394>

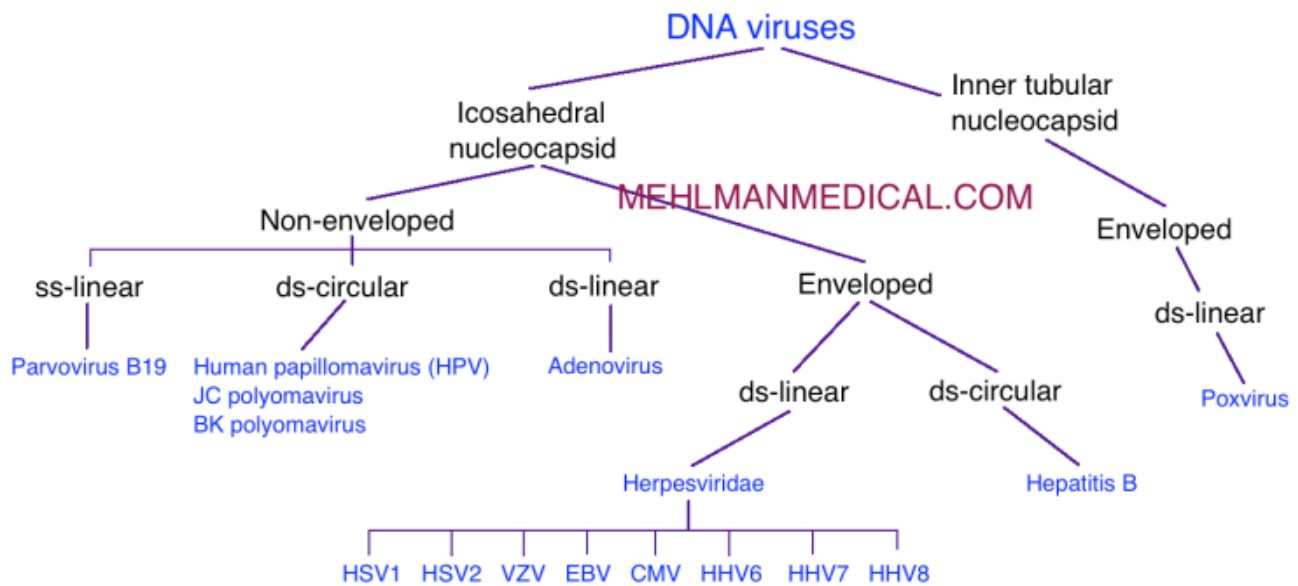
5) <https://cancerres.aacrjournals.org/content/69/1/6>

- 6) <https://www.ncbi.nlm.nih.gov/pubmed/12378375/>
- 7) <https://www.ncbi.nlm.nih.gov/pubmed/9606176/>
- 8) <https://www.ncbi.nlm.nih.gov/pubmed/10923632/>
- 9) <https://www.ncbi.nlm.nih.gov/pubmed/3243043/>
- 10) <https://www.ncbi.nlm.nih.gov/pubmed/10923631/>
- 11) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2570378/>

6. A 40-year-old IV drug user is admitted to hospital after experiencing a first-time focal epileptic seizure. He has a two-day history of headache, confusion, sensory aphasia, and right-sided hemiparesis. CD4 count is 135/ μ L. He is on highly active antiretroviral therapy (HAART). Other medications include trimethoprim-sulfamethoxazole (TMP-SMX). CSF analysis detects a DNA virus. Which of the following is most likely to be seen in this patient?

- A) Multifocal subcortical hyperintense lesions on T2-weighted MRI
- B) Temporal lobe atrophy with temporopolar grey/white matter abnormalities on T2-weighted MRI
- C) Ring enhancing lesions on T1-weighted MRI with gadolinium
- D) Hyperintensities in the striatum or thalamus on T2-weighted MRI

The answer is A.



This patient has **progressive multifocal leukoencephalopathy (PML)** caused by **JC polyomavirus**.

PML is a CNS infection characterized by destruction of oligodendrocytes and their myelin processes.¹ It is classically seen in HIV patients with CD4 T cell counts <300 cells/mm³.²

The predominant neurologic manifestations are disturbance of coordination, cognitive defects, and limb paresis.³

It is often described as causing **multifocal hyperintense lesions on T2-weighted MRI**. It causes hypointense lesions on T1-weighted MRI.^{4, 5, 6}

Yes, this sounds absurdly pedantic, but this description for whatever magical reason is HY on the USMLE.

“Temporal lobe atrophy with temporopolar grey/white matter abnormalities on T2-weighted MRI” (usually called “blurring”) refers to temporal lobe epilepsy.⁷ You do **not** need to know this for the USMLE.

“Ring enhancing lesions on T1-weighted MRI with gadolinium” refers to toxoplasmosis.⁸ However even if the vignette omitted telling you that a DNA virus was detected in the CSF, the fact that the patient is on TMP-SMX makes toxoplasmosis an absolute wrong answer on the USMLE since TMP-SMX is the prophylaxis for toxoplasmosis (and *Pneumocystis jirovecii*).⁹

“Hyperintensities in the striatum or thalamus on T2-weighted MRI” refers to Creutzfeldt-Jakob, which is a prion disease.¹⁰ MRI is classic for T2/FLAIR hyperintensities within the basal ganglia, thalamus, and cortex.¹¹ You do **not** need to know this for the USMLE.

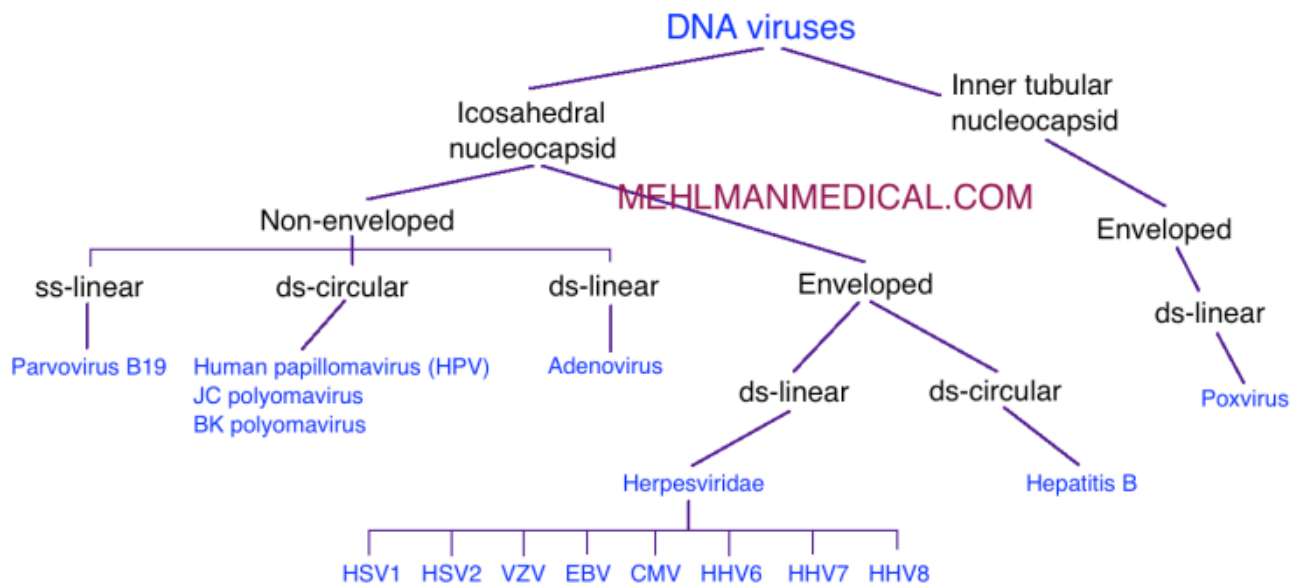
Bottom line: Progressive multifocal leukoencephalopathy (PML) is caused by JC polyomavirus. It is classically seen in HIV patients with CD4 counts <300 cells/mm³. Imaging shows multifocal hyperintense lesions on T2-weighted MRI.

- 1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5228722/>
- 2) <https://academic.oup.com/cid/article/31/4/e20/378128>
- 3) <https://academic.oup.com/jid/article/199/1/77/919695>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4495627/>
- 5) <http://www.ajnr.org/content/31/9/1564>
- 6) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6111648/>
- 7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3639571/>
- 8) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5483375/>
- 9) https://ascopubs.org/doi/abs/10.1200/JCO.2018.36.30_suppl.138
- 10) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5392034/>
- 11) <https://radiopaedia.org/articles/creutzfeldt-jakob-disease>

7. A researcher is doing a study on a virus specifically associated with renal transplants. It demonstrates an organ tropism largely limited to the reno-urinary tract and is known to cause a tubulointerstitial inflammatory response, ureteral stenosis, and hemorrhagic cystitis. Which of the following best describes the virus the researcher is studying?

- A) DNA; enveloped; linear
- B) DNA; enveloped; circular
- C) DNA; non-enveloped; linear
- D) DNA; non-enveloped; circular
- E) RNA; enveloped; linear
- F) RNA; enveloped; circular
- G) RNA; non-enveloped; linear
- H) RNA; non-enveloped; circular

The answer is D.



The researcher is studying **BK polyomavirus**, which is known to cause severe nephropathy, ureteral stenosis, and hemorrhagic cystitis in renal transplant recipients.^{1,2} This is the result of a tubulointerstitial inflammatory response similar to acute rejection.³

It is estimated that approximately 75% of adults are seropositive for BK polyomavirus but are asymptomatic⁴; immunocompromised hosts may experience infections, albeit they are rare overall outside of renal transplant recipients.³

It must be pointed out that cytomegalovirus (CMV) remains the most common opportunistic pathogen overall in organ transplant recipients^{5,6,7} (exceedingly HY on the USMLE), *however* CMV can affect almost any organ system, whereas BK polyomavirus demonstrates an organ tropism largely limited to the reno-urinary tract.⁸

Bottom line: BK polyomavirus can cause severe nephropathy, ureteral stenosis, and hemorrhagic cystitis after renal transplant. CMV remains the most common opportunistic pathogen overall following solid organ transplants, however it can cause infection in nearly any organ system; BK polyomavirus demonstrates an organ tropism largely limited to the reno-urinary tract.

1) <https://www.ncbi.nlm.nih.gov/pubmed/205555/>

2) <https://www.ncbi.nlm.nih.gov/pubmed/17493226>

3) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5084638/>

4) <https://www.ncbi.nlm.nih.gov/pubmed/20791873/>

5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4496754/>

- 6) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3848521/>
- 7) <https://onlinelibrary.wiley.com/doi/abs/10.1111/ctr.13512>
- 8) <https://www.ncbi.nlm.nih.gov/pubmed/17890268>

8.



A 42-year-old male presents with two-year history of a diffusely erythematous scaly rash over the trunk and extremities. He describes the rash as so intensely pruritic that he bathes several times a day in cold water to relieve the burning. On physical examination, he has multiple erythrodermic patches over his face and body. Abdomen is soft, non-tender, with no organomegaly. There is no lymphadenopathy. His vitals are within normal limits. A complete blood count is as follows:

Hb: 14 g/dL (NR for males: 13.5-17.5)

WBC: 23,000/mm³ (NR 4000-11,000)

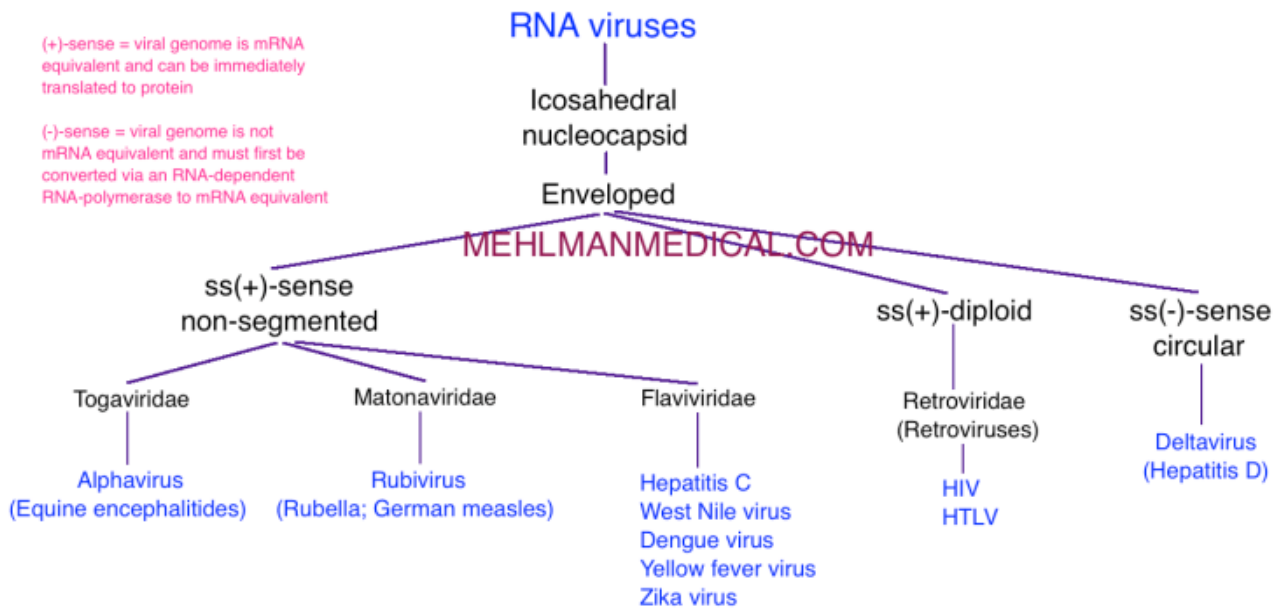
- Polymorphonuclear cells: ~10%

- Lymphocytes: ~90%

Peripheral smear examination reveals large lymphoid cells with prominent nuclear indentations and grooving. Bone marrow examination is unremarkable. Transmission electron microscopy (TEM) reveals characteristic cells with cerebriform nuclei. Which of the following is the most likely diagnosis?

- A) Sezary syndrome
- B) Mycosis fungoides
- C) HIV/AIDS
- D) Kaposi sarcoma
- E) Lichen planus

The answer is A.



Sezary syndrome is a form of low-grade cutaneous T cell non-Hodgkin lymphoma **with leukemic extension to the peripheral blood**; the purely cutaneous form of the disease is known as mycosis fungoides.¹ In other words, the syndrome is understood to be the advancing leukemic phase of mycosis fungoides.²

Sezary syndrome is characterized by the presence of “fiery red” erythroderma, generalized lymphadenopathy (in the early stages), and the presence of Sezary cells in the skin, lymph nodes, and peripheral blood.^{3,4} Sezary cells are described as “cerebriform” for their prominent nuclear indentations and grooving.⁵

The development of mycosis fungoides and Sezary syndrome may be caused by human T cell lymphotropic virus (HTLV).^{6,7}

- “Here we report the presence of HTLV pol and/or tax proviral sequences in 46 out of 50 (92%) of the patients tested. In addition, five of the patients, who lacked antibodies to HTLV-I/II structural proteins, were found to be seropositive for tax. It thus seems reasonable to conclude that MF/Sézary syndrome is an HTLV-associated disease and that lack of an immune response does not preclude infection with this type of virus.”⁶
- “The human T-cell lymphotropic virus type I (HTLV-I) is a retrovirus that causes adult T-cell leukemia–lymphoma, which can also involve the skin and be difficult to distinguish from a primary cutaneous T-cell lymphoma.”⁸

The USMLE also wants you to know that HTLV is endemic to Japan, with a contemporaneous increased prevalence of adult T cell leukemia and lymphoma.⁹ HTLV is also associated with IV drug use.¹⁰ Their contraction via IV drug use should not be a surprise as HTLV and HIV are both retroviridae, and HIV is frequently acquired via IV drug use. There have also been studies in the United States demonstrating the relationship of IV drug use to HTLV and HIV.¹¹

Kaposi sarcoma is a neoplasm of lymphatic endothelium infected with Kaposi sarcoma-associated herpesvirus (HHV8). It is characterized by the expression of lymphatic lineage-specific genes by the neoplastic Kaposi sarcoma cells.¹² Kaposi sarcoma is classically associated with AIDS and elderly men of Ashkenazi Jewish descent.¹³ The clinical appearance starts as violaceous macules that progress to plaques, papules, then nodules.¹⁴

Lichen planus is a skin condition characterized by the 6Ps – purple, pruritic, polygonal, planar, papules, and plaques. It is associated with hepatitis C infection.¹⁵

Bottom line: Mycosis fungoides and Sezary syndrome are cutaneous T cell non-Hodgkin lymphomas. MG is limited to the skin as a cutaneous T cell lymphoma; SS is the T cell leukemic extension. Both are characterized by cells with cerebriform nuclei. HTLV can cause MG and SS.

- 1) <https://www.sciencedirect.com/science/article/pii/B0122270800016189>
- 2) <https://www.ncbi.nlm.nih.gov/pubmed/12645344>
- 3) <https://www.sciencedirect.com/science/article/pii/B9781416045793100117>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5531705/>
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5531248/>
- 6) <https://www.ncbi.nlm.nih.gov/pubmed/7860737>
- 7) <https://www.ncbi.nlm.nih.gov/pubmed/9557617>
- 8) <https://www.nejm.org/doi/full/10.1056/NEJM199701233360415>
- 9) <https://www.ncbi.nlm.nih.gov/pubmed/22170555>
- 10) <https://www.ncbi.nlm.nih.gov/pubmed/2655084>
- 11) <https://www.ncbi.nlm.nih.gov/pubmed/2373871>
- 12) <https://www.ncbi.nlm.nih.gov/pubmed/15220917>
- 13) <https://www.ncbi.nlm.nih.gov/pubmed/18219366>
- 14) <https://www.sciencedirect.com/science/article/pii/B9781437729306000641>
- 15) <https://www.aafp.org/afp/2011/0701/p53.html>

9. An 18-year-old man presents to hospital with fever, increasing confusion, and aphasia. A CT scan of the head is unremarkable. EEG shows temporal complexes and patterns of slowing. A lumbar puncture is performed. Cerebral spinal fluid (CSF) findings are as follows:

Glucose: 70 mg/dL (normal range 50-80)

Protein: 75 mg/dL (normal range 15-60)

WBC: 220 cells/mm³; lymphocyte predominance

RBC: 45 cells/mm³

Opening pressure: 180 mm H₂O

Which of the following is the most likely diagnosis?

- A) Flavivirus
- B) Rhinovirus
- C) Echovirus
- D) Herpes simplex virus (HSV)
- E) Hanta virus

The answer is D.

This first chart is relatively qualitative and sufficient for the USMLE. The second chart is more quantitative/expansive in case you're interested; all values are heavily cited from the literature.

MEHLMANMEDICAL.COM					
Cerebral spinal fluid (CSF) findings as per notable CNS infections					
	Glucose (mg/dL)	Protein (mg/dL)	WBC type (cells/mm ³)	RBCs	Opening pressure (mm H ₂ O)
Normal CSF	50 to 80	15 to 60	Mononuclear (0-5)	None	60-200
Bacterial	↓	↑	↑ Neutrophils (PMNs)	None	↑
Viral (aseptic)	Normal	Normal or ↑	↑ Lymphocytes	None	Usually normal
Fungal	↓	Normal or ↑	↑ Lymphocytes	None	↑ (HY on USMLE!)
TB	↓	↑	↑ Lymphocytes	None	Normal or ↑
HSV	Normal or ↓	↑	↑ Lymphocytes	Usually ↑ (HY on USMLE!)	Normal or ↑

Cerebral spinal fluid (CSF) findings as per notable CNS infections						
MEHLMANMEDICAL.COM	Glucose (mg/dL)	Protein (mg/dL)	Predominant WBC cell type	WBC count (cells/mm ³)	RBCs (cells/mm ³)	Opening pressure (mm H ₂ O)
Normal CSF	50 to 80	15 to 60	Mononuclear	0-5	None	60-200
Bacterial	↓	↑ (100-500)	Neutrophils (Polymorphonuclears; PMNs)	>1,000	None	↑
Viral (aseptic)	Normal	Normal or ↑ (<95)	Lymphocytes	<100	None	Usually normal
Fungal	↓	Normal or ↑ (20-150)	Lymphocytes	10-500	None	↑ (HY on USMLE!)
Tuberculosis	↓	↑ (100-500)	Lymphocytes	100-500	None	Normal or ↑
Herpes simplex virus	Normal or ↓	↑ (>60)	Lymphocytes	>100-250	Usually ↑ (HY on USMLE!)	Normal or ↑

Opening pressure

- 10-100 mm H₂O in young children; 60 to 200 mm H₂O age >8; up to 250 mm H₂O in obesity.¹
- Fungal opening pressure increases proportional to fungal burden.^{2,3} Even though ↑ opening pressure is not specific for fungal meningitis, this association is HY on the USMLE.
- Opening pressure in HSV encephalitis is usually normal or increased.⁴

- In TB, opening pressure is $\uparrow$ roughly half the time.⁵
- In bacterial meningitis, opening pressure is >200 mm H₂O in 90% of cases and >500 mm H₂O in 15% of cases.⁶
- In viral, opening pressure is usually normal.⁷

Red blood cells (RBCs)

- Normal CSF should not have RBCs.⁷
- Unusual finding in CSF infections,⁷ and not characteristic of echovirus aseptic meningitis.⁸
- **Presence of RBCs ($>4-49/\text{mm}^3$) in an infective setting is highly characteristic of HSV encephalitis.**^{9, 10}

WBC count

- Normal range is $0-5/\text{mm}^3$.¹¹
- $>1,000$ per mm^3 in bacterial and <100 per mm^3 in viral.⁷
- $10-500$ cells/ mm^3 in fungal.¹²
- Usually $100-500$ cells/ mm^3 in tuberculosis¹³, although other studies have shown means of 71 cells/ mm^3 .¹⁴
- HSV usually ranges $>100-250$ cells/ mm^3 .¹⁵

WBC type

- Normal CSF has mononuclear predominance at $0-5/\text{mm}^3$.¹¹
- Neutrophils predominate in bacterial meningitis.⁷
- Lymphocytes predominate in viral, fungal, tubercular, and herpetic.^{7, 13, 15}

Protein

- Normal range is $15-60$ mg/dL.¹¹
- Bacterial is usually $100-500$ mg/dL.¹⁶
- Viral is usually <95 mg/dL.¹⁷
- Fungal is usually $20-150$ mg/dL.¹⁸
- TB is usually $100-500$ mg/dL.¹³
- For HSV, roughly half of patients are $60-100$ mg/dL of protein; the other half has >100 mg/dL.¹⁹

Glucose

- Normal range is $50-80$ mg/dL.⁷
- Bacterial is <40 mg/dL in 60% of patients.²⁰
- Normal for viral.²⁰
- Decreased in fungal and tuberculous meningitis.²¹
- Normal or decreased in HSV.^{22, 23}

Meningitis is an infection of the meninges, the tissue membranes surrounding the brain and spinal cord. Encephalitis is inflammation of the brain parenchyma itself.²⁴

This patient has **herpes encephalitis**, a hemorrhagic necrotizing encephalitis involving the frontal and medial temporal lobes.²⁵

Temporal lobe hemorrhage may be detected with CT or MRI²⁶, although imaging may indeed be negative.²⁷ EEG may show temporal complexes or patterns of slowing.^{28, 29}

Blood in the CSF is highly characteristic.^{9, 10}

Bottom line: HSV encephalitis is characterized by blood in the CSF due to temporal lobe hemorrhage. This may be detectable with CT or MRI, but imaging may indeed be negative. EEG may show temporal complexes or patterns of slowing. **This question is exceedingly HY on the USMLE.**

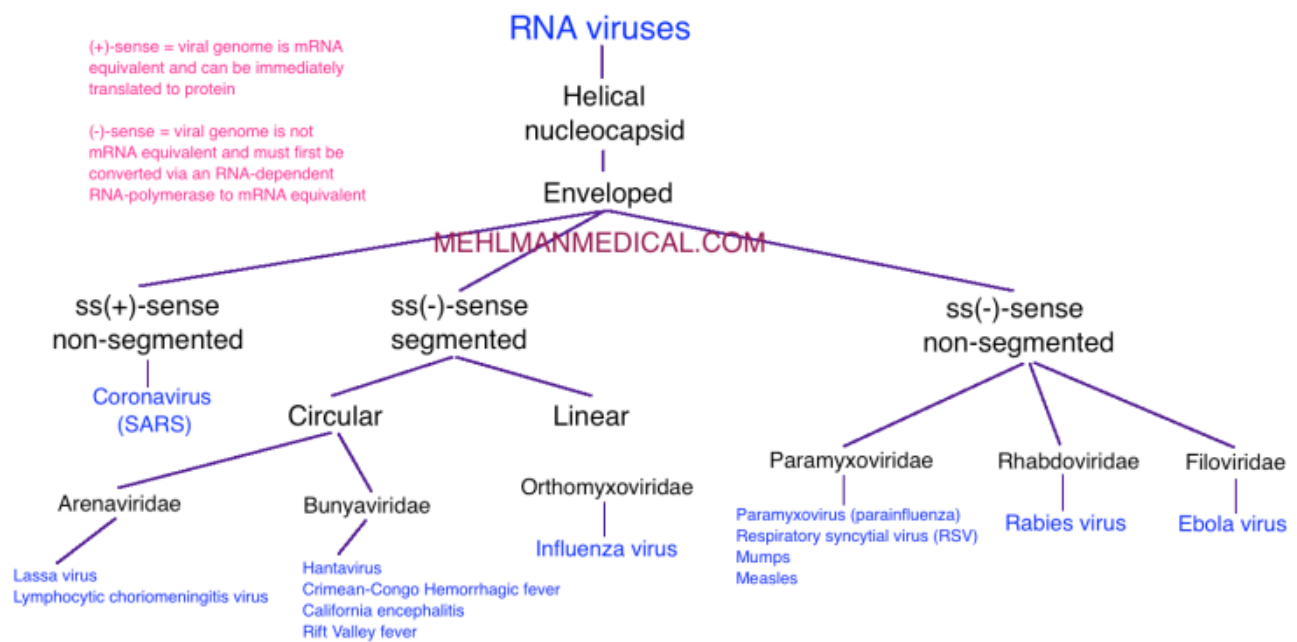
- 1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC493316/>
- 2) <https://academic.oup.com/cid/article/30/1/47/323550>
- 3) <https://www.ncbi.nlm.nih.gov/pubmed/19279443>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4965403/>
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2292571/>
- 6) https://books.google.co.jp/books/about/Cerebrospinal_Fluid_in_Clinical_Practice.html?id=o6SoJYPnyTQC&redir_esc=y
- 7) <https://www.aafp.org/afp/2003/0915/p1103.html>
- 8) [https://www.ijidonline.com/article/S1201-9712\(18\)30023-7/pdf](https://www.ijidonline.com/article/S1201-9712(18)30023-7/pdf)
- 9) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3560250/>
- 10) <https://bmcinfectdis.biomedcentral.com/articles/10.1186/1471-2334-12-356>
- 11) <https://medlineplus.gov/ency/article/003428.htm>
- 12) <https://emedicine.medscape.com/article/2172226>
- 13) <https://www.hindawi.com/journals/trt/2011/798764/>
- 14) <https://www.ncbi.nlm.nih.gov/pubmed/12921156>
- 15) <http://www.jpathinformatics.org/article.asp?issn=2153-3539>

- 16) <https://www.ncbi.nlm.nih.gov/books/NBK398/>
- 17) https://www.eaneurology.org/EFNS_guideline_2011_Routine_CSF_analysis.pdf
- 18) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4535692/>
- 19) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4753680/>
- 20) <https://www.medscape.com/answers/232915-10853/>
- 21) <https://www.sciencedirect.com/science/article/pii/B9780323077385100055>
- 22) <https://www.labcorp.com/tests/002048/glucose-cerebrospinal-fluid>
- 23) <https://www.ncbi.nlm.nih.gov/pubmed/3990441>
- 24) <https://www.ninds.nih.gov/Disorders/All-Disorders/Meningitis-and-Encephalitis-Information-Page>
- 25) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5244048/>
- 26) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5625746/>
- 27) <https://www.sciencedirect.com/science/article/pii/S1873959811000032>
- 28) <https://www.ncbi.nlm.nih.gov/pubmed/9637944>
- 29) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4828570/>

10. A 12-year-old girl has a three-day history of fever, sore throat, chills, myalgias, and general malaise. Her condition may have been prevented by receiving a yearly vaccine designed to induce an antibody response to a protein capable of binding sialic acid residues. Which of the following best describes the causative virus?

- A) RNA; non-enveloped; positive-sense; non-segmented
- B) RNA; non-enveloped; positive-sense; segmented
- C) RNA; non-enveloped; negative-sense; non-segmented
- D) RNA; non-enveloped; negative-sense; segmented
- E) RNA; enveloped; positive-sense; non-segmented
- F) RNA; enveloped; positive-sense; segmented
- G) RNA; enveloped; negative-sense; non-segmented
- H) RNA; enveloped; negative-sense; segmented

The answer is H.



This child has an influenza viral infection.

Two of the proteins encoded by the segmental genome, **hemagglutinin** and **neuraminidase**, are embedded in the viral envelope and available for antibody binding.¹

Influenza viral infection begins when **hemagglutinin binds to sialic acid residues on carbohydrate side-chains of cell-surface glycoproteins and glycolipids.**²

Following viral replication, neuraminidase cleaves sialic acid residues from the cell so that newly formed viral particles don't attempt to reinfect the same cell.^{3, 4}

Influenza vaccine

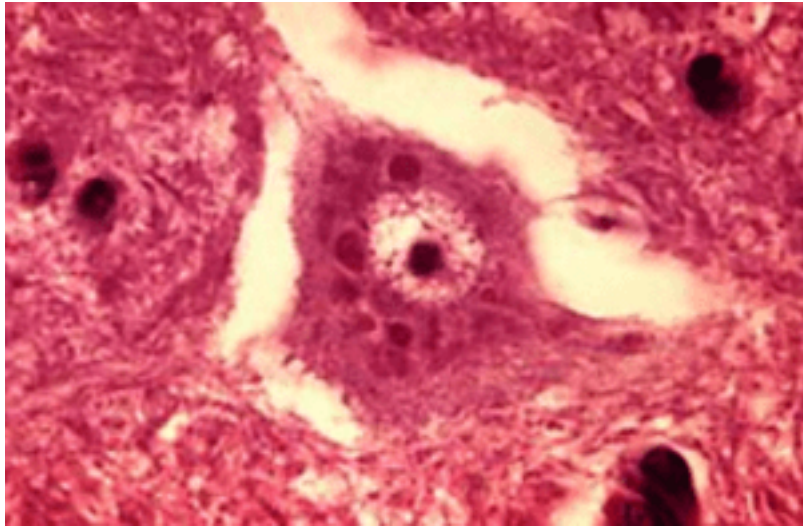
- Currently licensed vaccines are designed to induce antibodies against hemagglutinin.⁵
- Available in two main forms: killed (intramuscular) and live-attenuated (intranasal spray).⁶
- Killed vaccine may be given to anyone >6 months of age, including pregnant women.⁷
- Live-attenuated may be given to any non-pregnant, non-immunocompromised persons age 2-49.^{7, 8, 9, 10}

Bottom line: Influenza is a segmented RNA virus. Current vaccines are designed to produce antibodies to hemagglutinin. Killed vaccine may be given to anyone >6 months of age, including pregnant women. Live-attenuated vaccine may be given to any non-pregnant, non-immunocompromised persons age 2-49.

1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6520700/>

- 2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2937864/>
- 3) <https://www.sciencedirect.com/science/article/abs/pii/S0065323303010040>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2937864/>
- 5) <https://mbio.asm.org/content/9/2/e02332-17>
- 6) <https://www.who.int/biologicals/vaccines/influenza/en/>
- 7) <https://www.cdc.gov/vaccines/parents/diseases/flu.html>
- 8) <https://www.cdc.gov/vaccines/hcp/vis/vis-statements/flulive.html>
- 9) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4494344/>
- 10) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4718365/>

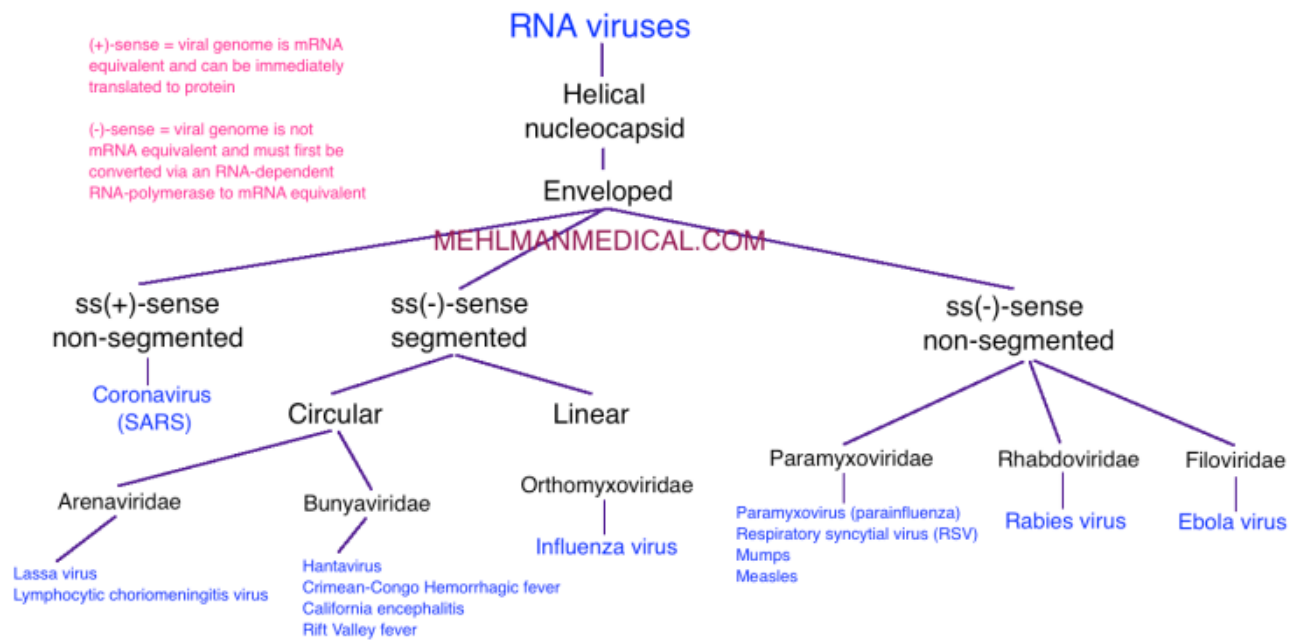
11.



A 15-year-old boy is biking through a wooded area in Vermont when he encounters a raccoon on the path. He runs it over, causing him to fall off his bike and scratch his hands. He proceeds to move the injured animal off the path and it bites him. Two months later, he experiences severe headache, paresthesia, and severe pain of his left hand. He presents to hospital febrile, with hypersalivation, hydrophobia, and aerophobia. He dies later the same day. Which of the following best describes the causative organism?

- A) RNA; non-enveloped; positive-sense; non-segmented
- B) RNA; non-enveloped; positive-sense; segmented
- C) RNA; non-enveloped; negative-sense; non-segmented
- D) RNA; non-enveloped; negative-sense; segmented
- E) RNA; enveloped; positive-sense; non-segmented
- F) RNA; enveloped; positive-sense; segmented
- G) RNA; enveloped; negative-sense; non-segmented
- H) RNA; enveloped; negative-sense; segmented

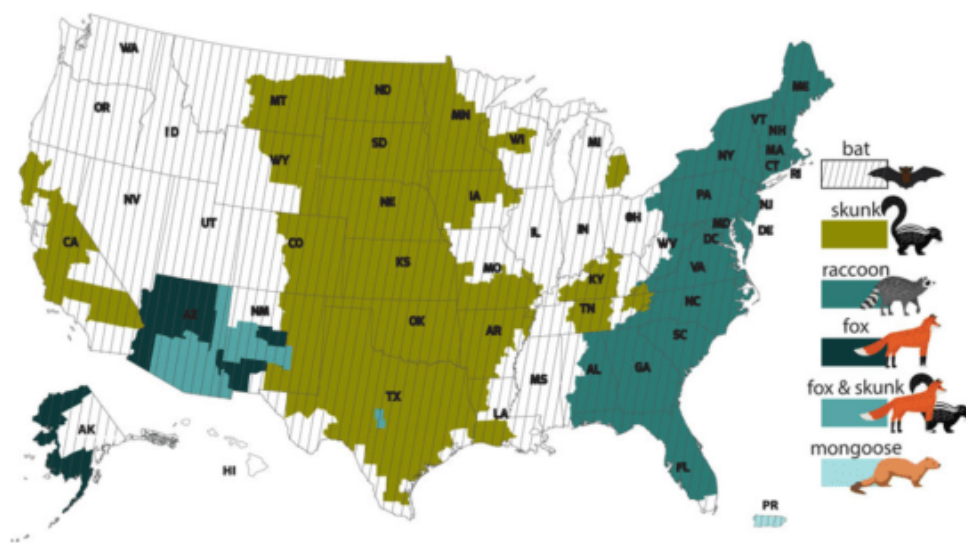
The answer is G.



This boy experienced a rabies infection.

The histological slide is of a **Negri body**, which is the classic histopathologic finding in rabies. Negri bodies are sharply demarcated, eosinophilic inclusions found in neurons most often in the hippocampal pyramidal cells, cerebellar Purkinje cells, and brainstem nuclei.¹

Rabies is a bullet-shaped virus with a tropism for neuronal tissue.² A bite from a wild animal is the most common route of transmission, however infections may also rarely be acquired through scratches, abrasions, or open wounds.³



Raccoons are the most common reservoir on the east coast of the United States.⁴

Following the bite / inoculation, the virus ascends at 12-14 mm/day to reach the spinal ganglion, where rapid viral multiplication takes place; it is during this rapid replication phase that paresthesias are experienced at the bite site. This is a pathognomonic feature of rabies and occurs in about half of cases. The virus then ascends at 200-400 mm/day to reach the brain and salivary glands. It binds to ACh, GABA, and glycine receptors. This is when encephalitic features present.⁵

The incubation period (i.e., time between contraction and first experiencing symptoms) for rabies is 20-60 days, with 1-3% of cases presenting >6 months later. Once symptoms start, within 2–10 days overt neurologic signs become apparent, ranging from hyperactivity to paralysis. The patient may experience confusion, agitation, **hydrophobia** (fear of drinking water because of painful swallowing), **aerophobia** (literally fear of air), **hypersalivation**, hyperventilation, priapism, and convulsions.⁶ Hydrophobia and aerophobia are pathognomonic of rabies.^{7, 8}

The USMLE Step 1 wants you to know vaccination information for post-exposure in unvaccinated individuals. The Step 2CK wants you to know all vaccination information:

Vaccination information⁹

- **Pre-exposure vaccination**

- For veterinarians, animal handlers, rabies laboratory workers, spelunkers, international travelers to endemic areas.
- Give 3 doses.
 - Day 0 – First dose
 - Day 7 – Second dose
 - Day 21 or 28 – Third dose

- **Post-exposure / bite**

- If previously not immune (unvaccinated)
 - Give **immunoglobulin + 4 doses of vaccine**.
 - Day 0 – Immunoglobulin + First dose of vaccine
 - Day 3 – Second dose
 - Day 7 – Third dose
 - Day 14 – Fourth dose
- If previously immune (vaccinated)
 - Do **not** give immunoglobulin. Give 2 doses of vaccine.
 - Day 0 – First dose of vaccine
 - Day 3 – Second dose

Bottom line: Rabies is a bullet-shaped RNA virus that forms Negri bodies in neuronal tissue. It is acquired most often through a bite from a wild animal. It has a long incubation period. Hydrophobia and aerophobia are pathognomonic. Vaccination information might seem pedantic but is sometimes assessed on the USMLE.

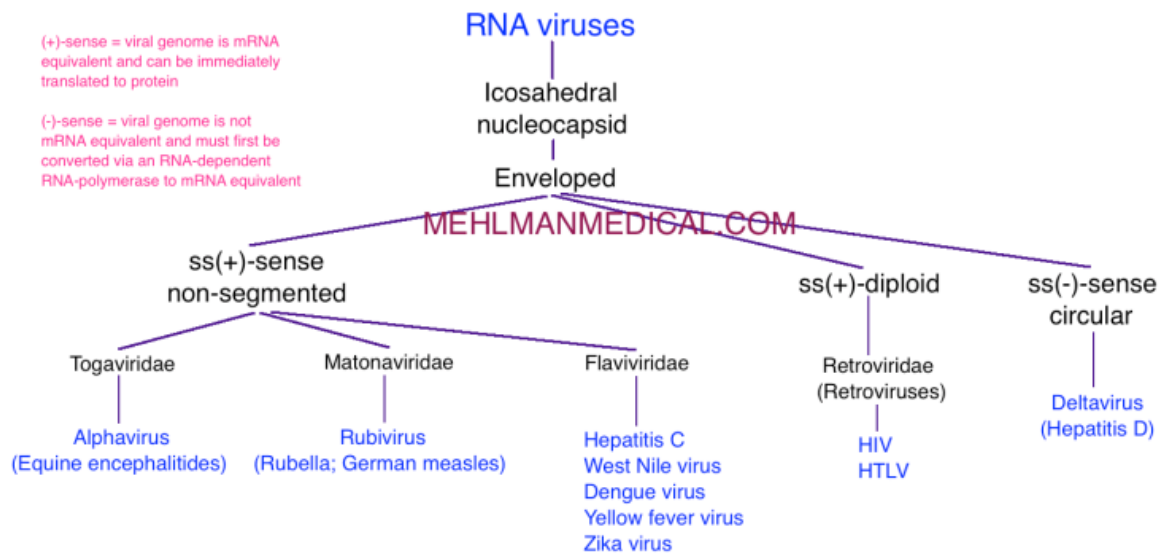
1) <https://www.sciencedirect.com/science/article/pii/B9780444534880000080>

- 2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4338442/>
- 3) <https://www.cdc.gov/rabies/transmission/index.html>
- 4) https://www.cdc.gov/rabies/location/usa/surveillance/wild_animals.html
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4338442/>
- 6) <https://academic.oup.com/cid/article/30/1/4/323391>
- 7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6293146/>
- 8) <https://link.springer.com/article/10.1007/BF02877781>
- 9) <https://www.cdc.gov/vaccines/hcp/vis/vis-statements/rabies.html>

12. A 30-year-old male presents to the physician for a one-week history of fever, generalized body pain, and vomiting. He appears pale and fatigued. A complete blood count is ordered and is unremarkable. He is recommended to take acetaminophen and drink copious fluids. One week later the fever persists. He reluctantly discloses to the physician that he recently tried IV drugs and had unprotected sex at a party. Hepatitis B, C, and HIV tests are ordered. Antibody testing comes back positive for hepatitis C. Polymerase chain reaction test (PCR) detects HCV RNA genotype 3. Which of the following is the most likely mechanism of hepatocellular injury in this patient?

- A) Direct viral cytotoxicity (Direct viral damage)
- B) Antibody-dependent cell-mediated cytotoxicity
- C) T cell-mediated hepatocellular apoptosis
- D) Th-inducing BTB/POZ domain-containing Kruppel-like zinc-finger transcription factor (ThPOK)

The answer is C.



Transmission of hepatitis C

- Hepatitis C is most frequently acquired via contaminated blood products or intravenous drug use.¹
- The risk of infection from a single needle-stick injury is 5-15% but may depend on the size of the inoculation.¹

Risk of transmission from sexual contact among heterosexual couples is low

- “Among 500 couples and 8377 person-years of sexual contact, the HCV Partners Study found a low prevalence of HCV infection among partners, with a maximum of 1.2% (95% CI: 0.2%–2.2%) of infections potentially attributable to sexual contact.”²
- “The maximal incidence of HCV infection was 7.2 per 10,000 person-years (95% CI: 1.3–13.0), and maximal risk per sexual contact was 1 per 380,000 (95% CI: 1/600,000–1/280,000).”²
- “Overall, available data indicate that HCV transmission by sex is reassuringly low, and the HCV Partners Study provides the numerical context to counsel HCV-infected persons in monogamous partnerships.”²
- “In contrast to hepatitis B, the risk of sexual transmission of HCV has always been considered low. This low risk was confirmed by a recent study among 500 anti-HCV-positive, HIV-negative persons and their long-term HCV-negative heterosexual partners, reporting a maximum incidence rate of HCV transmission by sex of 0.07% per year or one infection per 190,000 sexual contact...”³

Hepatocellular injury due to hepatitis C likely not due to direct viral damage

- “Although the exact mechanisms inducing the hepatic injury in chronic hepatitis C infection are still not completely elucidated, the evidence of a direct cytopathic viral effect is sparse. The overall majority of experimental studies and clinical observations among patients instead favor immune-mediated hepatocellular damage.”⁴

- “In immune competent hosts, there is little evidence that direct cytopathicity plays a significant role in liver cell injury.”⁵

Hepatocellular injury due to hepatitis C is believed to be T cell-mediated

- “The host immune response, in its attempt to clear the virus from the liver, contributes to the hepatocellular damage (chronic hepatitis) seen in the majority of chronically infected patients.”⁵
- “In HCV infection, upregulation of CD95 in hepatocytes as well as induction of CD95L expression in T-lymphocytes have been found to correlate with the severity of inflammation.”⁶
- “The prevalence of CD95 in HCV antigen-positive hepatocytes was significantly higher than in uninfected cells. Moreover, it has been suggested that the degree of CD95 expression correlates with hepatocyte apoptosis.”⁶
- “It has been proposed that CD95L-positive T-lymphocytes interact with CD95-bearing hepatocytes which results in liver cell apoptosis.”⁶
- “Cytotoxic T lymphocyte-mediated lysis of virus-infected host cells may lead to clearance of the virus or, if incomplete, to viral persistence and eventually chronic tissue injury.”⁷

Hepatocellular injury due to hepatitis A, B, and E is due to immune-mediated damage. Hepatitis D may cause direct viral cytotoxic effects in addition to heralding immune-mediated damage.^{8, 9, 10, 11, 12, 13}

Regarding hepatitis A

- “In particular, severe CD8+ T cell-mediated liver injury is known as a primary cause of tissue damage in hepatitis A virus (HAV) infection.”⁸

Regarding hepatitis B

- “It is generally accepted that the hepatitis B virus is not cytopathic and that liver cell damage in chronic HBV infection is dependent on the host’s immune response directed at viral and self-antigens on the surface of infected hepatocytes.”⁹
- “In these patients, both T and non-T cells from peripheral blood have been shown to be cytotoxic to autologous hepatocytes.”⁹

Regarding hepatitis D

- “It has been postulated that hepatocyte injury resulting from infection with hepatitis D virus may be caused by a direct virus cytotoxicity in contrast to immune-mediated injury associated with hepatitis B virus.”¹⁰
- “Hepatic cell death occurs due to direct cytotoxic effects of hepatitis D virus or a host-mediated immune response.”¹¹
- “HDV may cause hepatic failure through direct cytotoxicity caused by the S-HDAg or inducing an exaggerated immune response leading to the destruction of hepatocytes by cytotoxic T cells.”¹²

Regarding hepatitis E

- “The liver damage induced by HEV infection may be immune-mediated by cytotoxic T cells and natural killer (NK) cells since HEV is not cytopathic.”¹³

Antibody-dependent cell-mediated cytotoxicity is an immunologic mechanism via which an Fc receptor-expressing cell can recognize and kill antibody-coated target cells expressing tumor- or pathogen-derived antigens on their surface.¹⁴

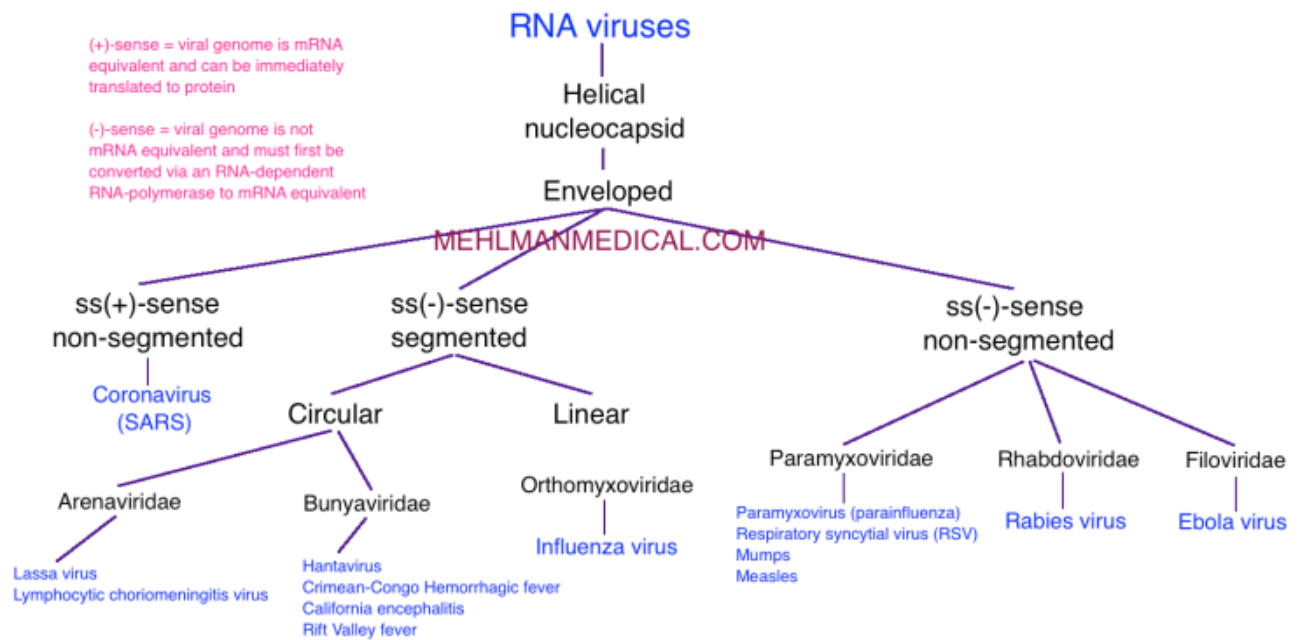
Bottom line: Hepatocellular injury due to the hepatitis viruses is overall most accurately described as **immune-mediated (usually T cell-)** rather than due to direct viral cytotoxic effects. Hepatitis D demonstrates contrast in that it may bear a lean toward the latter mechanism.

- 1) <https://www.ncbi.nlm.nih.gov/pubmed/7493305>
- 2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4659344/>
- 3) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6715947/>
- 4) <https://www.ncbi.nlm.nih.gov/pubmed/7481520>
- 5) <https://www.ncbi.nlm.nih.gov/pubmed/15560055>
- 6) <https://www.nature.com/articles/4401119>
- 7) <https://aasldpubs.onlinelibrary.wiley.com/doi/pdf/10.1002/hep.510300312>
- 8) https://www.jimmunol.org/content/192/1_Supplement/185.11
- 9) <https://www.ncbi.nlm.nih.gov/pubmed/3496375>
- 10) <https://www.ncbi.nlm.nih.gov/pubmed/1709411>
- 11) <https://www.ncbi.nlm.nih.gov/books/NBK470436/>
- 12) <https://academic.oup.com/gastro/article/7/4/231/5522133>
- 13) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4997574/>
- 14) <https://www.sciencedirect.com/science/article/pii/B9780123948021000017>

13. A 4-month old girl is brought to the GP by her mother for a two-day history of fever of 101F. On examination she appears healthy. There is no rash present. Soft expiratory crackles are auscultated diffusely across both lung fields. What is the most appropriate treatment for this patient?

- A) Ribavirin
- B) Palivizumab
- C) Motavizumab
- D) Inhaled corticosteroid
- E) Oral corticosteroid
- F) Antibiotic therapy
- G) Supportive care

The answer is G.



Respiratory syncytial virus (RSV) bronchiolitis is the most common cause of bronchiolitis in infants and young children and accounts for approximately 125,000 hospitalizations and 250 infant deaths every year in the United States. Peak incidence occurs at 2-3 months of age.¹ RSV is responsible for 50%–90% of children hospitalized for bronchiolitis.² Approximately two-thirds of infants are infected with RSV during the first year of life, and 90% have been infected one or more times by 2 years of age.³

Treatment is largely supportive (i.e., fluid replenishment and oxygen if needed).^{4, 5}

Ribavirin

Ribavirin, an RNA polymerase inhibitor, is the only FDA-approved treatment for RSV.^{5, 6} However the current (as of 2019) recommendation is that it should **not** be routinely used due to a combination of its cost and adverse effect profile.⁷

- “Ribavirin is licensed for use as a treatment for infants with bronchiolitis. However, the most recent Cochrane review of its use in 2007 suggested that the available evidence at that time was insufficient to confidently state whether or not ribavirin is clinically effective at treating RSV bronchiolitis. Therefore, due to its side-effect profile and lack of reproducible data on efficacy it is not currently recommended for routine use in RSV bronchiolitis in infants in the UK or US. The inhaled form of the drug is also extremely costly with wholesale prices for a single day of treatment quoted at US\$29,000 in the US.”⁷

Palivizumab and motavizumab

- Both monoclonal antibodies against RSV F-protein.⁷

- Palivizumab is sometimes as **prophylaxis, never as treatment**.⁵
- Indications for Palivizumab⁵
 - Preterm neonates <35 weeks gestation
 - Neonates with congenital heart or lung disease
- Motavizumab is 20x more potent than palivizumab but is not recommended for use due to lack of safety and clinical efficacy data.⁵

Corticosteroids, both inhaled and oral, are not indicated in the treatment of RSV.⁵

Bottom line: RSV is the most common cause of bronchiolitis in infants and young children; peak incidence is at 2-3 months of age. **Treatment is supportive care only.** Ribavirin is not frequently used as treatment and is the wrong answer on the USMLE. Palivizumab is used rarely as prophylaxis, not treatment, in neonates born <35 weeks gestation or who have congenital heart or lung disease. Motavizumab is a newer and more potent version of palivizumab but is not indicated for use due to lack of safety and clinical efficacy data.

1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5029757/>

2) <https://www.ncbi.nlm.nih.gov/pubmed/14980267/>

3) <https://www.ncbi.nlm.nih.gov/pubmed/3706232/>

4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2721379/>

5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4008286/>

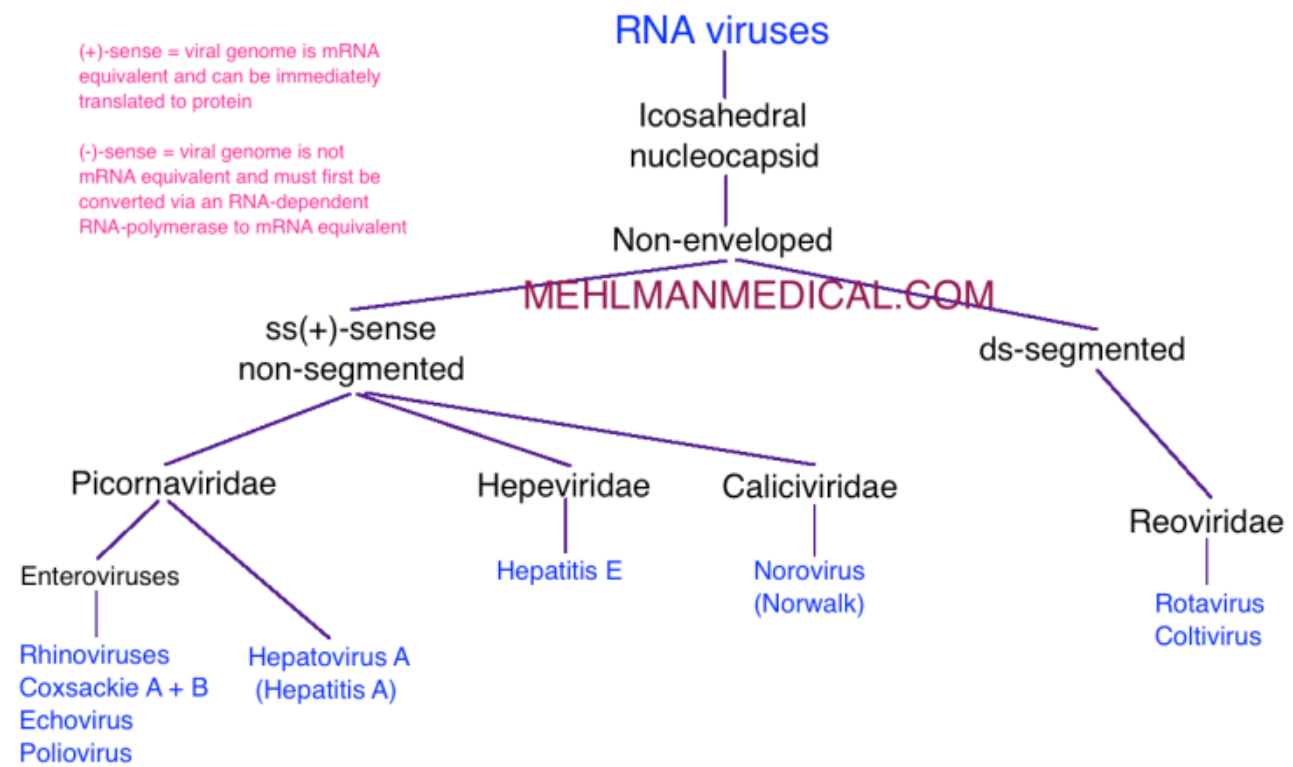
6) http://natap.org/hep_C/nm1200_1375.pdf

7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6664627/>

14. A 26-year-old woman at 34 weeks gestation presents to hospital with a two-day history of severe fatigue, loss of appetite, and abdominal pain. She is febrile at 102F. Blood pressure is 90/60 mm Hg. Physical examination shows scleral icterus. Abdomen is soft. There is voluntary guarding. Liver edge is palpable 4 cm below the right costal margin. Her history is significant for having returned to the United States from India two weeks ago. She is admitted to hospital for further investigation and her condition continues to deteriorate. She dies the following morning. Which of the following is the most likely causative pathogen?

- A) RNA; enveloped; single-stranded; non-segmented
- B) RNA; enveloped; single-stranded; segmented
- C) RNA; enveloped; double-stranded; non-segmented
- D) RNA; enveloped; double-stranded; segmented
- E) RNA; non-enveloped; single-stranded; non-segmented
- F) RNA; non-enveloped; single-stranded; segmented
- G) RNA; non-enveloped; double-stranded; non-segmented
- H) RNA; non-enveloped; double-stranded; segmented
- I) DNA; enveloped; linear
- J) DNA; enveloped; circular
- K) DNA; non-enveloped; linear
- L) DNA; non-enveloped; circular

The answer is E.



Hepatitis E has an incubation period of 2-10 weeks.¹ It is highly prevalent in South Asia, especially India, and is known to cause fulminant hepatitis and death in pregnant women.^{2,3}

Hepatitis E genotype 1 is associated with the greatest mortality in pregnant women, which may be as high as 15% to 25% of cases.⁴

HEV infection is especially severe during the second and third trimesters, with an increased number of obstetric complications, such as premature rupture of membranes (PROM), postpartum hemorrhage, spontaneous abortion, and intra-uterine fetal demise.⁵

It is believed that the hormonal changes (i.e., of estrogen and progesterone) during pregnancy are responsible for altering immune regulation and increasing the rate of viral replication. These immunologic changes cause downregulation of the p65 component of NF- κ B and a shift toward a predominantly CD4⁺ Th2-mediated response, which in turn may lead to fulminant illness in those with HLA susceptibility.⁶ CD4⁺ Th2-mediated responses do not typically assist in the clearance of viral illness⁷; CD8⁺ T cells are required for robust response to hepatitis E.⁸

Bottom line: Hepatitis E is associated with devastating outcomes in pregnant women, with high rates of stillbirth and fulminant hepatitis resulting in maternal death. The mechanism is related to pregnancy-induced hormonal changes that downregulate NF- κ B and shift the T cell response away from viral-clearing CD8⁺ T cells to CD4⁺ Th2 cells.

1) <https://www.who.int/news-room/fact-sheets/detail/hepatitis-e>

- 2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4501603/>
- 3) <http://www.ijmm.org/article.asp?issn=0255-0857>
- 4) <https://www.ncbi.nlm.nih.gov/pubmed/12490422/>
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4145510/>
- 6) <https://www.ncbi.nlm.nih.gov/pubmed/18662274/>
- 7) <https://www.ncbi.nlm.nih.gov/pubmed/10579123>
- 8) <https://www.ncbi.nlm.nih.gov/pubmed/31317546>

15. A 16-year-old girl has had increasing frontal headaches over the past four days that have become unbearable and associated with fevers and rigor. She reports having vomited once, as well as neck pain/stiffness and photophobia. A lumbar puncture is performed. Her cerebral spinal fluid (CSF) findings are as follows:

Glucose: 55 mg/dL (normal range 50-80)

Protein: 65 mg/dL (normal range 15-60)

WBC: 90 cells/mm³; lymphocyte predominance

RBC: 0-1 cells/mm³

Opening pressure: 180 mm H₂O

Which of the following is the most likely diagnosis?

- A) Herpes simplex virus (HSV)
- B) Rhinovirus
- C) Echovirus
- D) Flavivirus
- E) Hanta virus

The answer is C.

This first chart is relatively qualitative and sufficient for the USMLE. The second chart is more quantitative/expansive in case you're interested; all values are heavily cited from the literature.

MEHLMANMEDICAL.COM					
Cerebral spinal fluid (CSF) findings as per notable CNS infections					
	Glucose (mg/dL)	Protein (mg/dL)	WBC type (cells/mm ³)	RBCs	Opening pressure (mm H ₂ O)
Normal CSF	50 to 80	15 to 60	Mononuclear (0-5)	None	60-200
Bacterial	↓	↑	↑ Neutrophils (PMNs)	None	↑
Viral (aseptic)	Normal	Normal or ↑	↑ Lymphocytes	None	Usually normal
Fungal	↓	Normal or ↑	↑ Lymphocytes	None	↑ (HY on USMLE!)
TB	↓	↑	↑ Lymphocytes	None	Normal or ↑
HSV	Normal or ↓	↑	↑ Lymphocytes	Usually ↑ (HY on USMLE!)	Normal or ↑

Cerebral spinal fluid (CSF) findings as per notable CNS infections						
MEHLMANMEDICAL.COM	Glucose (mg/dL)	Protein (mg/dL)	Predominant WBC cell type	WBC count (cells/mm ³)	RBCs (cells/mm ³)	Opening pressure (mm H ₂ O)
Normal CSF	50 to 80	15 to 60	Mononuclear	0-5	None	60-200
Bacterial	↓	↑ (100-500)	Neutrophils (Polymorphonucleocytes: PMNs)	>1,000	None	↑
Viral (aseptic)	Normal	Normal or ↑ (<95)	Lymphocytes	<100	None	Usually normal
Fungal	↓	Normal or ↑ (20-150)	Lymphocytes	10-500	None	↑ (HY on USMLE!)
Tuberculosis	↓	↑ (100-500)	Lymphocytes	100-500	None	Normal or ↑
Herpes simplex virus	Normal or ↓	↑ (>60)	Lymphocytes	>100-250	Usually ↑ (HY on USMLE!)	Normal or ↑

Opening pressure

- 10-100 mm H₂O in young children; 60 to 200 mm H₂O age >8; up to 250 mm H₂O in obesity.¹
- Fungal opening pressure increases proportional to fungal burden.^{2,3} Even though ↑ opening pressure is not specific for fungal meningitis, this association is HY on the USMLE.
- Opening pressure in HSV encephalitis is usually normal or increased.⁴
- In TB, opening pressure is ↑ roughly half the time.⁵
- In bacterial meningitis, opening pressure is >200 mm H₂O in 90% of cases and >500 mm H₂O in 15% of cases.⁶

- In viral, opening pressure is usually normal.⁷

Red blood cells (RBCs)

- Normal CSF should not have RBCs.⁷
- Unusual finding in CSF infections,⁷ and not characteristic of echovirus aseptic meningitis.⁸
- Presence of RBCs ($>4-49/\text{mm}^3$) in an infective setting is highly characteristic of HSV encephalitis.^{9, 10}

WBC count

- Normal range is $0-5/\text{mm}^3$.¹¹
- $>1,000$ per mm^3 in bacterial and <100 per mm^3 in viral.⁷
- $10-500$ cells/ mm^3 in fungal.¹²
- Usually $100-500$ cells/ mm^3 in tuberculosis¹³, although other studies have shown means of 71 cells/ mm^3 .¹⁴
- HSV usually ranges $>100-250$ cells/ mm^3 .¹⁵

WBC type

- Normal CSF has mononuclear predominance at $0-5/\text{mm}^3$.¹¹
- Neutrophils predominate in bacterial meningitis.⁷
- Lymphocytes predominate in viral, fungal, tubercular, and herpetic.^{7, 13, 15}

Protein

- Normal range is $15-60$ mg/dL.¹¹
- Bacterial is usually $100-500$ mg/dL.¹⁶
- Viral is usually <95 mg/dL.¹⁷
- Fungal is usually $20-150$ mg/dL.¹⁸
- TB is usually $100-500$ mg/dL.¹³
- For HSV, roughly half of patients are $60-100$ mg/dL of protein; the other half has >100 mg/dL.¹⁹

Glucose

- Normal range is $50-80$ mg/dL.⁷
- Bacterial is <40 mg/dL in 60% of patients.²⁰
- Normal for viral.²⁰
- Decreased in fungal and tuberculous meningitis.²¹
- Normal or decreased in HSV.^{22, 23}

Meningitis is an infection of the meninges, the tissue membranes surrounding the brain and spinal cord. Encephalitis is inflammation of the brain parenchyma itself.²⁴

This patient has viral (aseptic meningitis). **Echovirus is the most common cause.**^{25, 26, 27}

Photophobia and neck stiffness are highly characteristic of meningitis. Aseptic is more common than bacterial.²⁸

Bottom line: Echovirus is the most common cause of viral meningitis. At a bare minimum, know the CSF findings for aseptic, bacterial, and fungal meningitis. RBCs in the CSF for HSV is also HY.

- 1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC493316/>
- 2) <https://academic.oup.com/cid/article/30/1/47/323550>
- 3) <https://www.ncbi.nlm.nih.gov/pubmed/19279443>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4965403/>
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2292571/>
- 6) https://books.google.co.jp/books/about/Cerebrospinal_Fluid_in_Clinical_Practice.html?id=o6SoJYPnyTQC&redir_esc=y
- 7) <https://www.aafp.org/afp/2003/0915/p1103.html>
- 8) [https://www.ijidonline.com/article/S1201-9712\(18\)30023-7/pdf](https://www.ijidonline.com/article/S1201-9712(18)30023-7/pdf)
- 9) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3560250/>
- 10) <https://bmcinfectdis.biomedcentral.com/articles/10.1186/1471-2334-12-356>
- 11) <https://medlineplus.gov/ency/article/003428.htm>
- 12) <https://emedicine.medscape.com/article/2172226>
- 13) <https://www.hindawi.com/journals/trt/2011/798764/>
- 14) <https://www.ncbi.nlm.nih.gov/pubmed/12921156>
- 15) <http://www.jpathinformatics.org/article.asp?issn=2153-3539>
- 16) <https://www.ncbi.nlm.nih.gov/books/NBK398/>
- 17) https://www.eaneurology.org/EFNS_guideline_2011_Routine_CSF_analysis.pdf

- 18) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4535692/>
- 19) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4753680/>
- 20) <https://www.medscape.com/answers/232915-10853/>
- 21) <https://www.sciencedirect.com/science/article/pii/B9780323077385100055>
- 22) <https://www.labcorp.com/tests/002048/glucose-cerebrospinal-fluid>
- 23) <https://www.ncbi.nlm.nih.gov/pubmed/3990441>
- 24) <https://www.ninds.nih.gov/Disorders/All-Disorders/Meningitis-and-Encephalitis-Information-Page>
- 25) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3298778/>
- 26) [https://www.ijidonline.com/article/S1201-9712\(18\)30023-7/pdf](https://www.ijidonline.com/article/S1201-9712(18)30023-7/pdf)
- 27) <https://www.ncbi.nlm.nih.gov/pubmed/3990441>
- 28) <https://www.ncbi.nlm.nih.gov/books/NBK545217/>

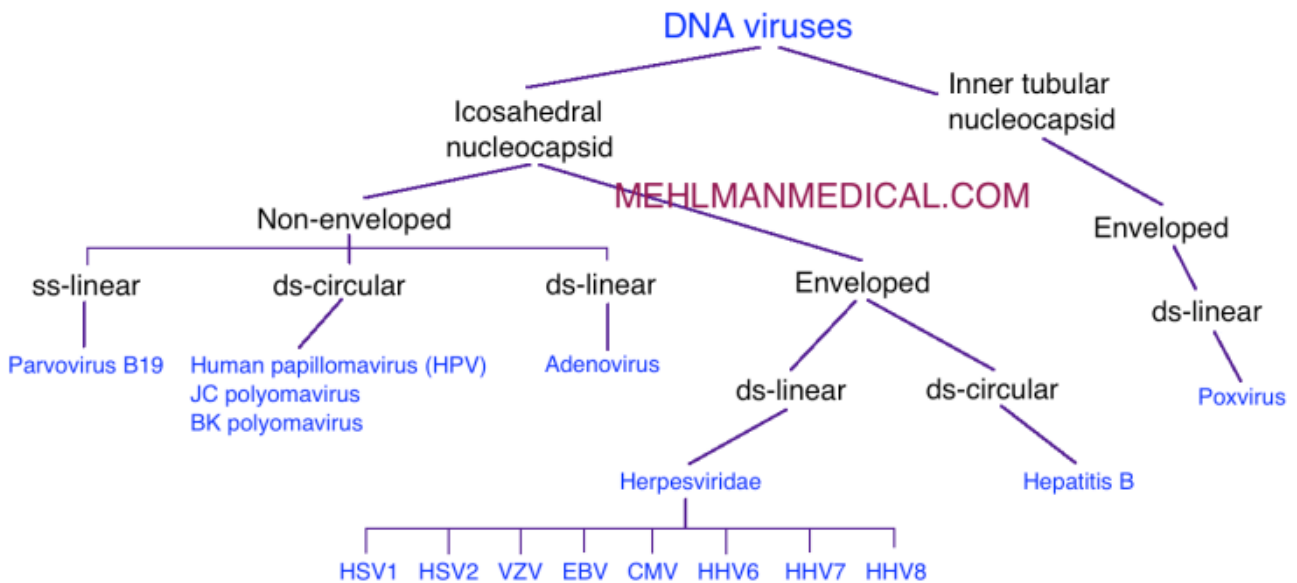
16.



A 6-year-old boy presents with the above rash. He has no prior dermatologic history. What is the most likely diagnosis?

- A) Herpetic whitlow
- B) Eczema herpeticum
- C) Shingles
- D) Mycosis fungoides
- E) Chickenpox

The answer is C.



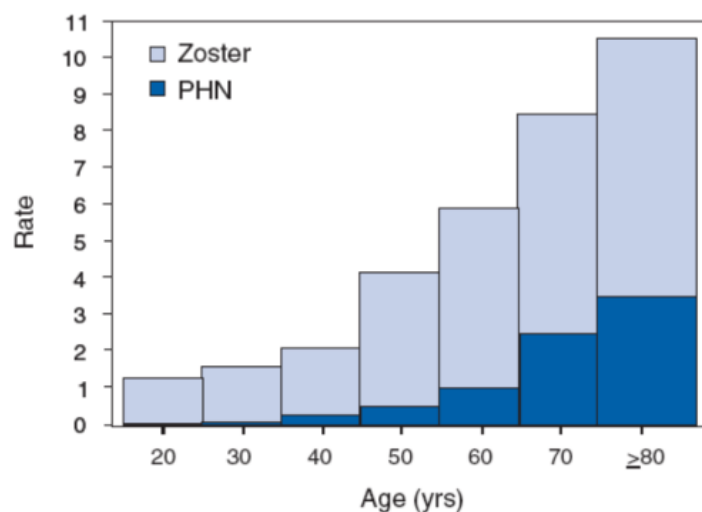
The diagnosis is **pediatric shingles**.

This is uncommon but HY for the USMLE.

In other words, after your initial what-the-fuck moment, you need to walk away from this question saying: “Okay, pediatric shingles is a thing. I know it exists now.”

Shingles is also known as herpes zoster, which is caused by varicella zoster virus (VZV), the same virus that causes chickenpox.¹

It is most common in older adults and immunocompromised individuals.^{2,3} One in three people will develop herpes zoster during their lifetime.⁴



Source⁵: <https://www.cdc.gov/shingles/surveillance.html>. Shingles and Postherpetic Neuralgia† Rates* by Age, United States. † Defined as pain for 30 days or longer. *per 1,000 person-years.

Super-HY point for the USMLE

- Herpes zoster is caused by reactivation of VZV from dorsal sensory or cranial nerve ganglia.⁵

Presentation

- There is a characteristic unilateral vesicular rash in a **dermatomal distribution** that consists of blisters that usually scab over in 7-10 days and evanesce within 2-4 weeks.^{6,7}
- Before the rash appears, people frequently have pain, itching, burning, or tingling in the area where it will develop for several days.⁶ This is called **preherpetic neuralgia**.⁸
- **Postherpetic neuralgia** is pain that persists following resolution of the rash.⁹

Pathogenesis

- Manifestation of shingles is closely linked to a reduction in VZV-specific cell-mediated immunity.¹⁰
- In patients with recent diagnosis, there is an observed decrease in natural killer (NK) cell activity, without a reduction in NK cell counts.¹¹
- The reduction in NK cell activity is characterized by a decrease in NK cell secretion of interferon-gamma (IFN- γ).¹⁰
- Frequent previous psychological stress events are increased compared to controls in those with recent diagnosis of shingles.¹⁰

Pediatric shingles

- Can occur at any time after VZV infection or vaccination.¹²
- Incidence is less after varicella vaccination compared to natural infection.¹²
- Infants with shingles have an observable reduction in levels of lymphocytes, NK cells, and virus-specific immunoglobulins. This may explain the lesser ability to maintain the latency of VZV.¹³

Diagnosis of shingles

- Usually diagnosed on clinical grounds only (i.e., history and physical exam)¹⁴
- If a diagnostic modality is used, a Tzanck smear of scrapings from the floor of the vesicles reveals multinucleated giant cells.¹⁵

Treatment for shingles

- **Oral Acyclovir**, valacyclovir, or famciclovir is the treatment.¹⁶
- IV is used for severely immunocompromised patients (i.e., transplant recipients, those on chemotherapy, etc.) with disseminated disease.^{16,17}
- Children under 12 generally do not receive antiviral therapy.¹²

Herpetic whitlow is HSV1 or 2 infection of a digit, usually seen as vesicles on the finger in those in the dental profession.¹⁸ It may be treated with acyclovir.¹⁹

Eczema herpeticum is an HSV1 or 2 superinfection on eczema. *This is an actual herpes infection* and is not merely herpetiform (herpes-like). It is treated with acyclovir.²⁰ This patient has no prior dermatologic history. Patients with eczema herpeticum would have a history of eczema.

Mycosis fungoides is a cutaneous T cell lymphoma usually caused by human T-cell lymphotropic virus.²¹ It has characteristic cerebriform cells.²²

Chickenpox is primary VZV infection. It presents as crops of vesicles at different stages of healing. The virus goes latent in sensory and cranial nerve ganglia and may reemerge later in life as herpes zoster.^{23, 24}

Bottom line: Shingles (herpes zoster) is caused by reactivation of varicella zoster virus from dorsal sensory or cranial nerve ganglia. Pathogenesis is linked to decreased VZV-specific cell-mediated immunity and NK cell activity. Oral acyclovir, valacyclovir, or famciclovir is the treatment. Although much more common with increasing age, pediatric shingles is a well-established condition in dermatology literature.

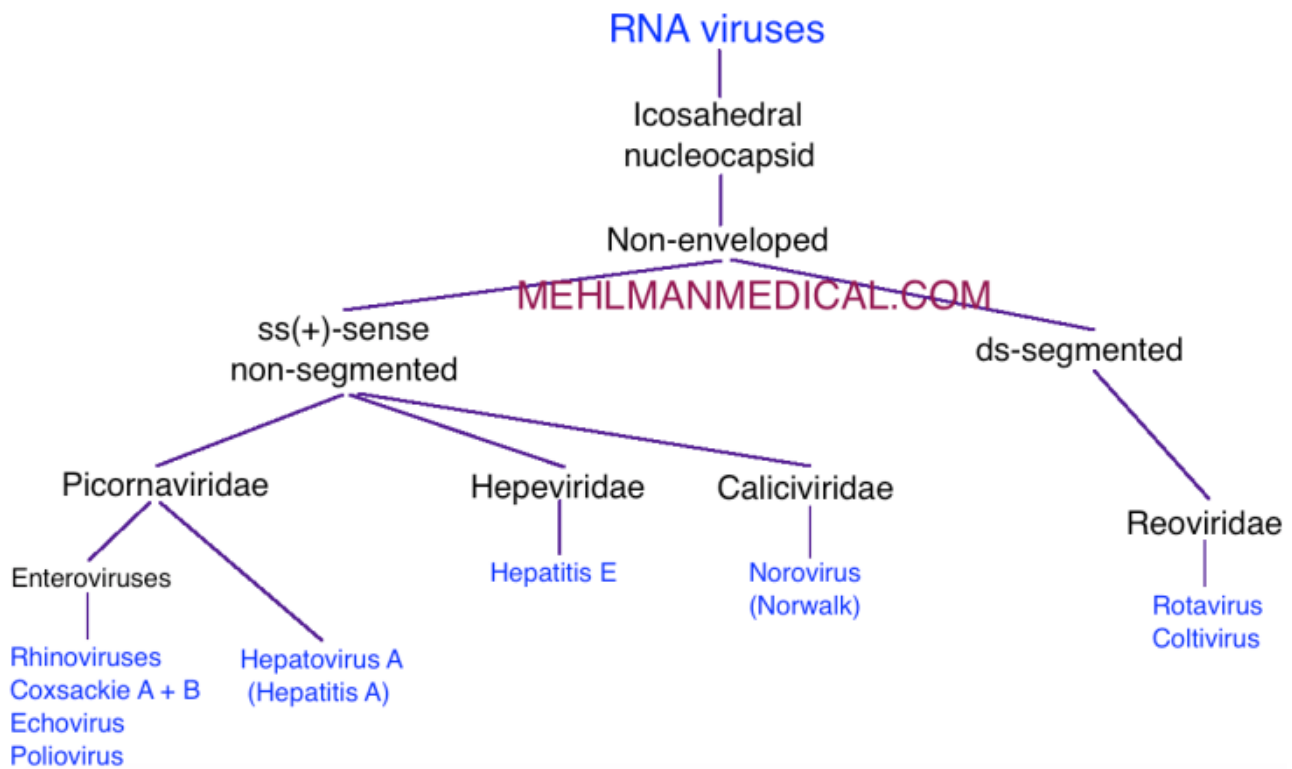
- 1) <https://www.cdc.gov/shingles/index.html>
- 2) <https://www.cdc.gov/mmwr/preview/mmwrhtml/rr5705a1.htm>
- 3) <https://www.ncbi.nlm.nih.gov/pubmed/17939900>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2664599/>
- 5) <https://www.cdc.gov/shingles/surveillance.html>
- 6) <https://www.sciencedirect.com/science/article/pii/B9780323029483000286>
- 7) <https://www.cdc.gov/shingles/about/symptoms.html>
- 8) <https://www.ncbi.nlm.nih.gov/pubmed/1866008>
- 9) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3479946/>
- 10) <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0193299>
- 11) <https://www.ncbi.nlm.nih.gov/pubmed/2978454>
- 12) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5029240/>

- 13) <https://www.ncbi.nlm.nih.gov/pubmed/16409922/>
- 14) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1934966/>
- 15) <https://www.ncbi.nlm.nih.gov/pubmed/2448353/>
- 16) <https://www.aafp.org/afp/2000/0415/p2437.html>
- 17) https://academic.oup.com/cid/article/44/Supplement_1/S1/334966
- 18) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3437498/>
- 19) <https://www.ncbi.nlm.nih.gov/pubmed/2957639>
- 20) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3520662/>
- 21) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC295510/>
- 22) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3699909/>
- 23) <https://www.ncbi.nlm.nih.gov/pubmed/6316159>
- 24) <https://www.sciencedirect.com/science/article/pii/B9780123737410500106>

17. A 3-year-old girl has a two-day history of dyspnea with feeding, vomiting, and pallor. On arrival, she is tachypneic, tachycardic, hypotensive, and febrile. Physical examination shows eyelid puffiness, hepatomegaly, and generalized edema. Her mother reports that she recently had a runny nose for a few days and a low-grade fever that self-resolved. The most likely pathogen responsible for this patient's condition is also known to be associated with which of the following?

- A) Renal insufficiency
- B) Hand-foot-mouth disease
- C) Herpangina
- D) Pulmonary hypertension
- D) Diabetes mellitus

The answer is E.



This patient has dilated cardiomyopathy caused by **coxsackie B virus**-induced myocarditis.

Myocarditis is a common cause of dilated cardiomyopathy.¹ The majority of cases of myocarditis are secondary to a viral infection, with the most common in children being coxsackie B virus.^{2,3}

The most likely pathophysiology is **direct viral cytotoxicity** to myocardium.⁴ This mechanism is exceedingly HY on the USMLE because it contrasts with rheumatic fever due to Group A Strep, which is a type II hypersensitivity.⁵

Dyspnea with feeding, vomiting, and pallor are frequently seen in infants with viral-induced cardiomyopathy. Tachypnea, tachycardia, and fever are also common clinical signs. Hypotension reflects a decompensation in cardiac function.⁶

In addition to dilated cardiomyopathy, coxsackie B virus is associated with the development of diabetes mellitus type I.⁷

Mechanism

- Viral induction of autoimmunity is believed to be due to either T cell activation or molecular mimicry. Coxsackie B4 virus is strongly associated with the development of diabetes mellitus type I and shares sequence homology with the islet auto-antigen glutamic acid decarboxylase (GAD).⁸

- An amino acid sequence of coxsackie B virus, called p2C, demonstrates homology with a sequence in islet auto-antigen GAD65 and binds to the diabetes associated HLA-DR3 molecule.⁹
- The major diabetes auto-antigen, GAD65, contains a region of sequence similarity, including six identical residues (*PEVKEK*) that bear homology with the p2C protein of coxsackie B virus. This cross-reactivity between coxsackie B virus and GAD65 can initiate diabetes mellitus type I.¹⁰

In other words, when the immune system mounts a response to the viral infection, it simultaneously attacks the beta islet cells.

Hand-foot-mouth disease and herpangina are due to coxsackie A, not B, virus.^{11, 12}

Hand-foot-mouth disease is most common under the age of five and is characterized by painful vesicles on the hands, feet, and mouth, as the name implies.¹³

Herpangina causes a painful enanthem that typically occurs on the soft palate, tonsils, and posterior oropharynx.¹¹

Bottom line: Coxsackie B virus can cause dilated cardiomyopathy and diabetes mellitus type I. The coxsackie B virus p2C protein bears a six-amino-acid oligopeptide sequence that demonstrates molecular mimicry with islet cell auto-antigen glutamic acid decarboxylase 65. When the immune system mounts a response to the viral infection, it simultaneously attacks the beta islet cells.

1) <https://www.ncbi.nlm.nih.gov/pubmed/18192257>

2) <https://www.ncbi.nlm.nih.gov/pubmed/4212998/>

3) <https://www.ncbi.nlm.nih.gov/pubmed/13570836/>

4) <https://www.ncbi.nlm.nih.gov/pubmed/14695668>

5) <https://books.google.co.jp/books?id=-4xbFuo6aQgC&pg=PA37>

6) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2805590/>

7) <http://www.imedpub.com/articles/role-of-coxsackie-virus-b-in-type-1-diabetesbrief-review.pdf>

8) <https://www.ncbi.nlm.nih.gov/pubmed/9662368>

9) <https://www.ncbi.nlm.nih.gov/pubmed/9498628>

- 10) <https://www.jimmunol.org/content/165/7/3830>
- 11) <https://www.ncbi.nlm.nih.gov/books/NBK507792/>
- 12) <https://www.ncbi.nlm.nih.gov/pubmed/22404077>
- 13) <https://www.cdc.gov/hand-foot-mouth/index.html>

18.



A 21-year-old woman presents with the above skin findings on her back. She says the lesions are itchy and have gradually increased in number over the past couple of weeks. She is sexually active and infrequently uses barrier contraception. Her palms and soles show no abnormalities. Which of the following is the most likely diagnosis?

- A) Atopic dermatitis
- B) Tinea versicolor
- C) Nummular eczema
- D) Seborrheic dermatitis
- E) Lichen planus
- F) Secondary syphilis
- G) Pityriasis rosea

The answer is G.

This spot diagnosis is exceedingly HY on the USMLE!

Pityriasis rosea is a self-limiting cutaneous eruption with a very characteristic presentation:

- Begins as a **herald patch**, or mother patch, which is an erythematous, round to oval, scaly patch or plaque, 2-10 cm in diameter, with a depressed center and raised border.¹



- This is followed by crops of smaller lesions that erupt on the trunk and proximal extremities in a “**Christmas tree**” distribution (i.e., the eruption on the trunk/back + proximal shoulders is apparently euphemized to resemble a Christmas tree).^{1, 2}
- The rash lasts 1-8 weeks in 85% of patients, with an average duration of 5 weeks, before self-resolving.³
- About half of cases are preceded by a non-specific upper respiratory tract-like presentation.⁴
- The rash is described as mildly to severely pruritic in 25% of patients.⁴
- Greatest incidence is 20-29 years of age.⁴

The exact etiology of pityriasis rosea may be manifold, however viral infection with human herpes virus 6 and 7 (HHV6/7), **particularly HHV7**, is believed to be highly associated.^{5, 6, 7, 8}

- “Although the detection of HHV7 DNA in peripheral blood mononuclear cells (PBMC) and tissues does not prove directly a causal role, HHV7 DNA in cell-free plasma corresponds to active replication which supports a causal relationship. We propose that pityriasis rosea is a clinical presentation of HHV7 reactivation.”⁵
- “Light and electron microscopy findings suggest infection with human herpesviruses 6 and 7 (HHV6/7). These viral antigens have been detected in skin lesions by immunohistochemistry and their DNA has been isolated from non-lesional skin, peripheral blood mononuclear cells, serum, and saliva samples.”⁶

- “Pityriasis rosea is an acute exanthem with many clinical and epidemiologic features of an infectious disease. To date, human herpesvirus (HHV)-6 and HHV-7 appear to be the most indicted culprits.”⁷
- “Nested polymerase chain reaction and *in situ* hybridization enabled detection of human herpesvirus-7 and human herpesvirus-6 in skin and other tissues isolated from patients with pityriasis rosea. These results suggest that pityriasis rosea is associated with systemic active infection with both human herpesvirus-7 and human herpesvirus-6.”⁸

Atopic dermatitis is the most common chronic skin disease in children, typically presenting <2 years of age. It is characterized by pruritic eczema flares and often seen as part of the atopy constellation (asthma, allergic rhinoconjunctivitis, eczema). Serum IgE and eosinophils may be elevated.⁹ Treatment is with emollients and topical steroids.¹⁰

Tinea versicolor is a cutaneous fungal infection caused by *Malassezia furfur*.¹¹ It is described as oval macules that coalesce into larger patches with a fine scale on the shoulders, chest, and back. The yeast blocks melanin synthesis in the skin and also produces a skin bleaching agent. With ultraviolet exposure, a hypopigmented spotted appearance is enhanced due to contrast with the darkened surrounding skin.¹² Treatment is typically with topical selenium.¹³

Nummular eczema is characterized by an itchy rash that forms coin-shaped patches on the skin. It is treated with topical steroids.¹⁴

Seborrheic dermatitis is a chronic inflammatory skin disorder often caused by the fungus *Pityrosporum ovale*. It is characterized by powdery or greasy scales in skin folds and along hair margins. Treatment is usually selenium or ketokonazole shampoo.¹⁵

Lichen planus is a skin condition characterized by the 6Ps – purple, pruritic, polygonal, planar, papules, and plaques). It is associated with hepatitis C infection.¹⁶

Secondary syphilis notably produces a body rash that includes the palms and soles. Treatment of syphilis is with penicillin.¹⁷ This patient’s palms and soles are unaffected.

Bottom line: Pityriasis rosea is a self-limiting cutaneous eruption classically manifesting as a herald patch followed by the emergence of a rash that spreads in a “Christmas tree” distribution. The exact etiology may be manifold, however viral infection with human herpes virus 6 and 7 (HHV6/7), **particularly HHV7**, is believed to be highly associated. This is an important spot diagnosis on the USMLE. Be familiar with the image.

1) <https://www.ncbi.nlm.nih.gov/books/NBK513301/>

2) <https://www.ncbi.nlm.nih.gov/books/NBK448091/>

3) <https://www.ncbi.nlm.nih.gov/pubmed/2595391>

- 4) <https://www.aafp.org/afp/2004/0101/p87.html>
- 5) <https://www.ncbi.nlm.nih.gov/pubmed/9529560>
- 6) <https://www.bmj.com/content/351/bmj.h5233>
- 7) <https://www.ncbi.nlm.nih.gov/pubmed/20797508>
- 8) <https://www.sciencedirect.com/science/article/pii/S0022202X15300324>
- 9) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4470205/>
- 10) <https://www.ncbi.nlm.nih.gov/pubmed/18208463>
- 11) <https://www.sciencedirect.com/topics/medicine-and-dentistry/tinea-versicolor>
- 12) <https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/tinea-versicolor>
- 13) <https://www.ncbi.nlm.nih.gov/books/NBK293716/>
- 14) <https://www.aocd.org/page/NummularEczema>
- 15) <https://www.aafp.org/afp/2000/0501/p2703.html>
- 16) <https://www.aafp.org/afp/2011/0701/p53.html>
- 17) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1071050/>

19. A 45-year-old man who recently arrived in the US from China has various screening tests performed as part of a routine immigration physical. His serum studies for hepatitis B are as follows:

Anti-HBcAb IgM: Positive

Anti-HBcAb IgG: Negative

HBsAg: Negative

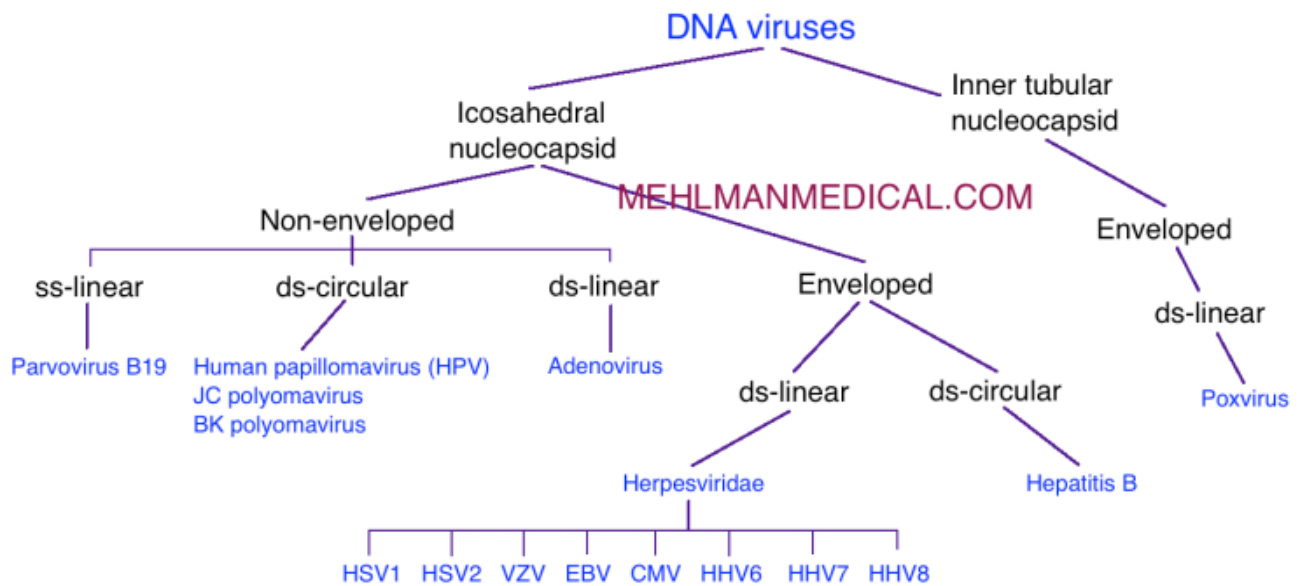
HBsAb: Negative

HBeAg: Negative

Which of the following is the most accurate interpretation of these findings?

- A) Recently vaccinated but not yet not immune
- B) Recently vaccinated with immunity conferred
- C) Recently vaccinated and in the window period
- D) Recently infected but immunity conferred
- E) Recently infected and in the window period

The answer is E.



In terms of understanding the window period, first let's quickly define the meaning of each serum marker:

- HBs – surface antibody¹
 - A marker of immunity.
 - **Positive if immune; negative if not immune.**
 - This should pretty much be the first marker you look at when interpreting HBV lab results. You can immediately just say, “Okay, the patient is immune.” Or, “Ok, he/she is not immune.”
- HBsAg – surface antigen¹
 - A marker of infectivity.
 - **If positive, indicates current infection, either acute or chronic.**
 - If (+), means the patient cannot be immune because he or she clearly has a current infection.
- HBeAg – “e” antigen¹
 - Marker of robust infectivity.
 - Correlates with a high level of viral replication.
 - **If positive, means person with HepB can easily transmit virus to a non-immune individual.**
 - Helps guide clinical management.
- HBcAb – core antibody¹
 - Marker of acute, chronic, or resolved HBV infection.
 - **If positive, means person has had the actual disease before, either now or in the past.**
 - Cannot be positive if the patient has never been infected.
 - **Negative if vaccinated.**
- HBcAb IgM
 - Subclass of HBcAb.¹
 - Positivity indicates recent infection with HBV (<6 mos).¹
 - **Positive during the window period.²**

- HBcAb IgG¹
 - Subclass of HBcAb.
 - Marker of past or current infection.
 - If HBsAg is also positive (and HBcAb IgM is negative), indicates chronic HepB infection.

As the patient begins to develop immunity to HepB, he or she will start producing HBsAb. At the same time, as he or she begins to clear the virus, HBsAg will decrease.

The short time period between when HBsAg has been reduced to undetectable levels, while simultaneously, HBsAb has not yet risen to detectable levels, denotes the window period.²

Because HBsAb is not yet positive, the patient is not yet considered immune to hepatitis B.

If the USMLE asks you which serum marker is most reliable during the window period, the answer is HBcAb IgM because it's the only one that's positive.

MEHLMANMEDICAL.COM						
Hepatitis B serology interpretation						
	HBsAg	HBsAb	HBcAb	HBcAb IgM	HBcAb IgG	HBeAg
Acute infection (low infectivity)	+	-	+	+	-	-
Acute infection (high infectivity)	+	-	+	+	-	+
Chronic infection (low infectivity)	+	-	+	-	+	-
Chronic infection (high infectivity)	+	-	+	-	+	+
Immune (previous infection) (<6 months)	-	+	+	+	-	-
Immune (previous infection) (>6 months)	-	+	+	-	+	-
Immune (vaccinated)	-	+	-	-	-	-
Window period	-	-	+	+	-	-
Susceptible (not immune)	-	-	-	-	-	-

Bottom line: Interpretation of hepatitis B serology is exceedingly HY. If the USMLE asks you which serum marker is most reliable during the window period, the answer is HBcAb IgM because it's the only one that's positive.

1) http://www.state.in.us/isdh/files/Hepatitis_B_Facts.pdf

2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5189866/>

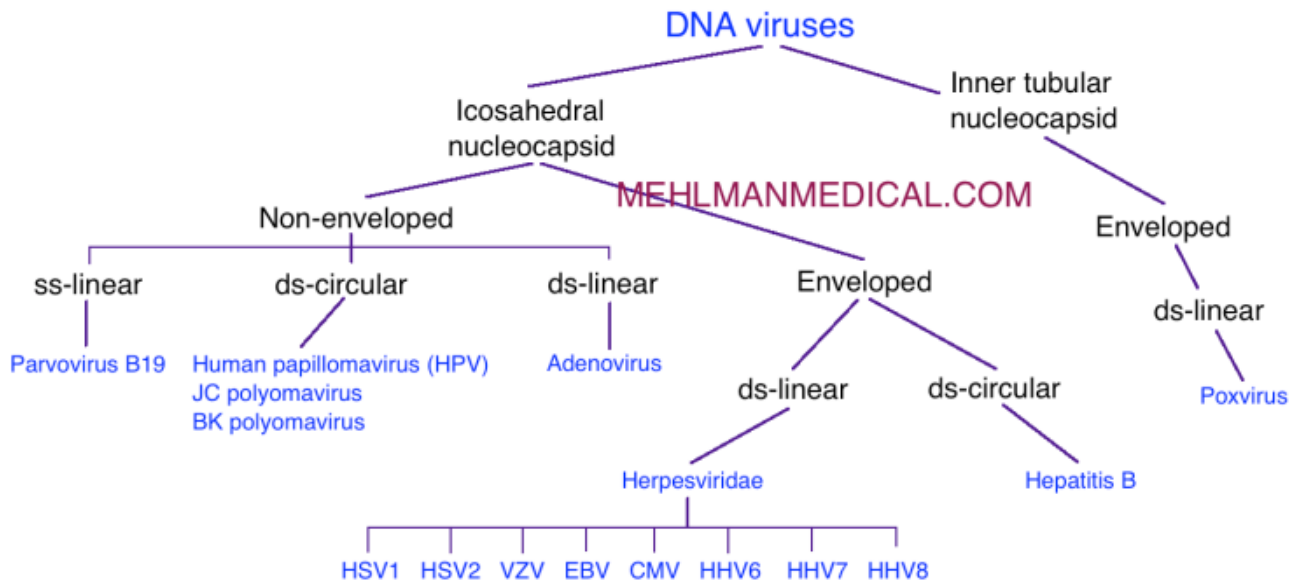
20.



A 52-year-old HIV-positive male who not taken his highly active antiretroviral therapy (HAART) for the past three months presents with new-onset violaceous lesions of his back, trunk, lower extremities. Which of the following best describes the histologic findings of these lesions?

- A) Basal cell vacuolization with band-like lymphocytic infiltrate at the dermo-epidermal junction with Civatte bodies
- B) Spindle cell proliferation of lymphatic endothelial cells forming sinuous vascular spaces, with PAS-positive intracellular and extracellular hyaline globules
- C) Parakeratosis and thickened projections of the prickle cell layer of keratinocytes with polymorphonuclear and lymphocytic infiltrate
- D) Epidermal spongiosis with elongation of intercellular bridges and leukocytic infiltration

The answer is B.



Kaposi sarcoma is a neoplasm of lymphatic endothelium infected with human herpes virus 8 (HHV8), also known as Kaposi sarcoma-associated herpesvirus (KSHV). It is characterized by the expression of lymphatic lineage-specific genes by the neoplastic Kaposi sarcoma cells.¹ Kaposi sarcoma is classically associated with AIDS and elderly men of Ashkenazi Jewish descent.² The clinical appearance starts as violaceous macules that progress to plaques, papules, and then nodules.³

Pathophysiology

HHV8 infection of B cells induces multiple cytokines and chemokines that can drive development of the cancer.⁴

- “HHV-8 targets a range of naive B cell, IgM memory B cell, and plasma cell-like populations for infection and induction of interleukin-6, tumor necrosis factor alpha, macrophage inhibitory protein 1α, macrophage inhibitory protein 1β, and interleukin-8 in vitro and in the blood of HHV-8/HIV-1-coinfected subjects with KS.”⁴
- “The histogenesis of Kaposi’s sarcoma tumor spindle cells remains controversial but several immunohistochemical studies favor a lymphatic origin.”⁵

Histology of Kaposi sarcoma

- “There may be varying degrees of overlying epidermal changes which can range from prominent hyperkeratosis and acanthosis to frank ulceration. The dermal proliferation is comprised of a spindle cell proliferation of endothelial cells forming sinuous vascular spaces. In plaque and nodular stage lesions, there may be visible intracellular and extracellular hyaline globules, thought to represent engulfed erythrocytes. While rare this may be seen in patch stage lesions. They stain PAS positive.”⁶

Lichen planus

“Basal cell vacuolization with band-like lymphocytic infiltrate at the dermo-epidermal junction with Civatte bodies” refers to lichen planus, which is a skin condition characterized by the 6Ps – purple, pruritic, polygonal, planar, papules, and plaques). It is associated with hepatitis C infection.⁷

- “The characteristic histopathological changes including basal cell vacuolization, band-like lymphocytic infiltrate at dermo-epidermal junction were seen in all the biopsies while Civatte bodies were detected in 29% cases. The CBs represent necrotic keratinocytes, measure about 20 µm in diameter and have a homogeneous, eosinophilic appearance.”⁸

Psoriasis

“Parakeratosis and thickened projections of the prickle cell layer of keratinocytes with polymorphonuclear and lymphocytic infiltrate” refers to psoriasis.

- “Histology of psoriasis is characterised by parakeratosis (cell nuclei within stratum corneum) and thickened projections of the prickle cell layer of keratinocytes (psoriasiform hyperplasia). There is no granular layer. Polymorphonuclear leukocytes and lymphocytes infiltrate dermis (CD8+) and epidermis (CD4+).”⁹

Eczema

“Epidermal spongiosis with elongation of intercellular bridges and leukocytic infiltration” refers to eczema.

- “The spongiotic tissue reaction pattern is characterised by intercellular oedema within the epidermis (spongiosis). Initially, there is a widening of intercellular spaces between keratinocytes and elongation of the intercellular bridges. Further accumulation of fluid leads to the formation of intraepidermal vesicles. Dermal changes include varying degrees of oedema and a superficial perivascular infiltrate with lymphocytes, histiocytes and occasional neutrophils and eosinophils.”¹⁰

Bottom line: Kaposi sarcoma is a neoplasm of lymphatic endothelium caused by HHV8 (KSHV). It is classically associated with AIDS and elderly men of Ashkenazi Jewish descent. It presents grossly as violaceous macules that progress to plaques, papules, and then nodules. Histologically there is spindle cell proliferation of lymphatic endothelial cells forming sinuous vascular spaces.

1) <https://www.ncbi.nlm.nih.gov/pubmed/15220917>

2) <https://www.ncbi.nlm.nih.gov/pubmed/18219366>

3) <https://www.sciencedirect.com/science/article/pii/B9781437729306000641>

- 4) <https://www.ncbi.nlm.nih.gov/pubmed/25182322>
- 5) <https://www.ncbi.nlm.nih.gov/pubmed/16615115>
- 6) <https://dermnetnz.org/topics/kaposi-sarcoma-pathology/>
- 7) <https://www.aafp.org/afp/2011/0701/p53.html>
- 8) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4037946/>
- 9) <https://dermnetnz.org/cme/scaly-rashes/psoriasis-overview/>
- 10) <https://dermnetnz.org/topics/eczema-pathology/>



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